

The coppe document class

Version 3.8

Vicente H. F. Batista George O. Ainsworth Jr.
Geraldo Xexéo Eduardo Mangeli

May 28, 2026

Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibT_EX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a mailing list where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain
`\mainmatter` three-parts: *front*, *main* and *back* matters [?]. Each one of these parts is started
`\backmatter` by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`.
The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (`capa`)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The `capa` is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation [?]. It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your
`\foreigntitle` monograph in the current and foreign languages. The default languages are Brazilian Portuguese and English. The `babel` package is automatically loaded by `coppe.cls`, so you do not need to load it again. The Brazilian Portuguese is the main language and the English is only required for the foreign abstract.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc.
`\examiner`

students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the Academic Registry [?], abstracts must be in one page each, with
`foreignabstract (env.)` at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.
`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.
`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which
`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is
`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for
`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_EX produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_T_EX `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_T_EX styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
```

```
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript's reference list):

Academic `@book` (`@livro`); `@article` (`@artigo`); `@artigodejornal` (newspaper); `@incollection` (`@partedelivro`); `@inbook` (`@emlivro`); `@inreference` (`@verbete`); `@proceedings` (`@anais`); `@inproceedings` (`@trabalhodeevento`); `@thesis` (`@tese/@dissertacao`, with the presets `@dissertacaomestrado`, `@tesedoutorado`, `@tesephD`, `@tcc`); `@report` (`@relatorio`); `@periodical` (`@periodico`); `@manual`; `@misc` (`@diverso`); `@online`.

Special (NBR 6023 §4.2.5–4.2.14) `@patent` (`@patente`); `@legislation` (`@legislacao/@atonormat`); `@jurisdiction` (`@jurisprudencia`, with a `relator` field); `@legal` (`@documentojuridico`); `@standard` (`@norma`); `@music` (`@partitura`); `@map` (`@mapa`); `@image` (`@iconografico`); `@artwork` (`@obradearte`); `@video` (`@audiovisual`); `@movie` (`@filme`); `@letter` (`@correspondencia`); `@dataset` (`@conjuntodedados`).

Games and TV `@game` (`@jogo`); `@boardgame` (`@jogodetabuleiro`); `@videogame` (`@jogoeletronico`); `@tvshow` (`@programadetv`).

When an entry has no `author` or `editor`, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPETEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

english COPPET_{EX} uses Babel. The default language is Portuguese (actually `brazilian`), with English being the second language. If option `english` is used, English becomes the main language and Portuguese the secondary. Look at the Babel package to switch between languages.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

6 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and `pdflatex` will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package. It is also useful to enable hyperlinks. Fortunately, the `coppe` class will do this automatically if `hyperref` is loaded.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

7 A simple example

```

1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{booktabs}% tabelas mais bonitas
5 \usepackage{rotating}% rodando coisas, como tabelas
6 \usepackage{longtable} % tabelas longas
7 \usepackage[most]{tcolorbox} % caixas de texto
8 \usepackage{tikz}          % gráficos vetoriais (o tcolorbox já o traz)
9 \usepackage{subcaption}    % subfiguras/subtabelas (NÃO use subfig/subfigure)
10 \usepackage{amsmath,amssymb}
11 \DeclareMathOperator{\RMSE}{RMSE}% operadores próprios (cap. de matemática)
12 \DeclareMathOperator{\diag}{diag}
13 \usepackage{hyperref}
14 % listings, fancyvrb e algorithm2e já vêm carregados pela classe coppe (v3.8).
15
16 \addbibresource{example.bib}% base de referências (biblatex/biber)
17
18 \makelossymbols
19 \makeloabbreviations
20
21 \begin{document}
22   \title{Título da Tese}
23   \foreigntitle{Thesis Title}
24   \author{Nome do Autor}{Sobrenome}
25   \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
26   \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
27   \advisor{Prof.}{Nome do Terceiro Orientador}{Sobrenome}{D.Sc.}
28
29   \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
30   \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
31   \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
32   \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
33   \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
34   \department{PESC}
35   \date{01}{2024}

```

```

36
37 \keyword{Primeira palavra-chave}
38 \keyword{Segunda palavra-chave}
39 \keyword{Terceira palavra-chave}
40
41 \maketitle
42
43 \frontmatter
44 \dedication{A alguém cujo valor é digno desta dedicatória.}
45
46 \chapter*{Agradecimentos}
47
48 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
49 orientadores, colegas, familiares e \as ag\ências de fomento. Lorem ipsum
50 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
51 ut labore et dolore magna aliqua.
52
53 \begin{abstract}
54
55 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
56 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
57 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
58 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
59 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
60 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
61 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
62 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
63 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
64 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
65 aspernatur aut odit aut fugit.
66
67 \end{abstract}
68
69 \begin{foreignabstract}
70
71 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
72 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
73 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
74 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
75 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
76 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
77 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
78 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
79 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
80 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
81 aspernatur aut odit aut fugit.
82
83 \end{foreignabstract}
84
85 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
86 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
87 \listoffigures
88 \listoftables
89 \listofquadros

```

```

90 \listofprogramas
91 \listofalgorithms
92 \printloabbreviations
93 \printlosymbols
94 \tableofcontents
95
96 \mainmatter
97 \chapter{Introdução}
98
99 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
100
101 A classe \verb|coppe| foi criada por Vicente Helano e George Ainsworth, porém, em 2024, é
102
103 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.com}
104
105 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
106
107 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
108
109 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
110
111 Apesar desse modelo ser muito bom, ele tem um defeito: a limitação do sistema de referênci
112
113 Mais ainda, as regras da COPPE ainda não se adaptaram, no início de 2024, as novas regras
114
115 Este documento não substitui, mas complementa, o documento que descreve a classe.
116
117
118
119 \chapter{Configurações Iniciais}
120
121 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
122
123 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
124
125 Recomendo ler o documento “The \verb*|coppe| document class” para entender melhor todas
126
127 \section{Linguagem principal do texto}
128
129 Essa classe considera que o texto principal está em português e algumas partes específicas
130 \begin{itemize}
131 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
132 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
133 \end{itemize}
134
135 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe \CoppeT
136
137
138 \section{Por que usar o \LaTeX}
139
140 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess
141
142 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p
143

```

144 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
145
146 \section{Como e onde usar o \LaTeX}
147
148 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambientes
149 \begin{enumerate}
150 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Windows
151 \item Usar um ambiente na rede, como o Overleaf.
152 \end{enumerate}
153
154 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o backup
155
156
157 \chapter{Algumas Regras da COPPE}
158
159 Todas abreviaturas e símbolos devem ser definidas antes de utilizadas. Isso é facilmente feito
160
161 É imprescindível definir os símbolos, tal como o
162 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
163 \syml{\mathbb{R}}{Conjunto dos números reais}
164 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
165
166 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \verb|makeindex|
167
168 Como as listas de símbolos e de abreviaturas usamos o mesmo comando usado para criar índices
169
170 \section{Citações}
171 \label{sec:citacoes}
172
173 Citações curtas podem ser feitas \code{\quote{...}} ou direto com “duas crases e dois pontos”
174
175 \begin{longquote}
176 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
177 feita com o ambiente \code{\verb=longquote=}: The primary objective of this
178 investigation was to determine the feasibility of detecting corrosion in
179 aluminum Naval aircraft components with neutron radiographic interrogation
180 and the use of standard corrosion penetrameters. Secondary objectives
181 included the determination of the effect of object thickness on image quality,
182 the defining of minimum levels of detectability and a preliminary investigation
183 of a means whereby the degree of corrosion could be quantified with neutron
184 radiographic data. \code{\cite{article-example}}
185 \end{longquote}
186
187 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \code{\verb*|natbib|}
188
189 Em todo caso, \code{\textbf{deve}} se tomar enorme atenção com as citações, para evitar ocorrer erros
190
191 Quando se cita uma obra por meio de outra (\code{\textit{apud}}), pela ABNT entra na lista de referências
192
193 \chapter{Floats}
194
195 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \code{\textit{floats}}
196
197 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesma

```

198
199 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração}
200
201 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao
202
203
204 \section{Tabelas e Figuras Padrão}
205
206 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
207
208 \begin{table}[ht]
209 \centering % Centraliza a tabela
210 \caption{Exemplo de Tabela de Números}
211 \label{tab:exemplo_numeros}
212 \begin{tabular}{ccc} % Define a quantidade de colunas
213 \hline % Linha superior
214 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
215 \hline % Linha média
216 1 & 2 & 3 \\ % Primeira linha de dados
217 \hline
218 4 & 5 & 6 \\ % Segunda linha de dados
219 \hline
220 7 & 8 & 9 \\ % Terceira linha de dados
221 \hline
222 10 & 11 & 12 \\ % Quarta linha de dados
223 \hline % Linha inferior
224 \end{tabular}
225 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
226 \end{table}
227
228
229
230 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
231
232 \begin{figure}[ht]
233 \centering % Centraliza a figura
234 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
235 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
236 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
237 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
238 \end{figure}
239
240 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
241
242 \begin{quadro}[ht]
243 \centering
244 \caption{Exemplo de Quadro}
245 \label{qua:exemplo}
246 \begin{tabular}{|l|l|}
247 \hline
248 \textbf{Ilustração} & \textbf{Delimitação} \\
249 \hline
250 Tabela & Linhas apenas no topo e na base. \\
251 \hline

```

```

252 Quadro & Linhas em todas as bordas. \\
253 \hline
254 \end{tabular}
255 \fonte{Elabora\c{c}\~ao pr\'opria.}
256 \end{quadro}
257
258
259
260
261
262 \section{Tabelas mais elegantes}
263
264 Atualmente a tend\^encia \e usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
265
266 \begin{table}[ht]
267 \centering % Centraliza a tabela
268 \caption{Exemplo de Tabela de N\^umeros mais elegantes}
269 \label{tab:exemplo_numerosbom}
270 \begin{tabular}{ccc} % Define a quantidade de colunas
271 \toprule % Linha superior
272 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \textbf{Cabe\c{c}alhos}
273 \midrule % Linha m\^edia
274 1 & 2 & 3 \\ % Primeira linha de dados
275 4 & 5 & 6 \\ % Segunda linha de dados
276 7 & 8 & 9 \\ % Terceira linha de dados
277 10 & 11 & 12 \\ % Quarta linha de dados
278 \bottomrule % Linha inferior
279 \end{tabular}
280 \fonte{Elabora\c{c}\~ao pr\'opria.}
281 \end{table}
282
283 \section{Tabelas Longas ou Largas}
284
285 Se sua tabela \e muito longa ou larga, existem v\arias op\c{c}o\~es.
286 \begin{itemize}
287 \item alterar o tamanho da letra
288 \item Usar o longtable
289 \item rodar a tabela, fazendo ela em \textit{landscape}
290 \item fazer a tabela dentro de um minibox
291 \end{itemize}
292
293
294 \subsection{Tabelas largas demais}
295
296 \E comum em teses que as tabelas sejam largas demais. H\aa v\arias formas de resolver isso.
297
298 A \autoref{tab:tabela_largafns} \e larga demais, e nela isso \e resolvido diminuindo a fonte p
299
300 \begin{table}[ht]
301 \centering % Centraliza a tabela
302 \caption{Exemplo de Tabela Larga com Fonte Menor}
303 \label{tab:tabela_largafns}
304 \footnotesize % Aplica uma fonte menor para a tabela
305 \begin{tabular}{ccccccc} % Aumente o n\^umero de colunas conforme necess\ario

```

```

306 \toprule
307 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Coluna 5} \\
308 \midrule
309 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
310 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
311 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
312 \bottomrule
313 \end{tabular}
314 \fonte{Elabora\c{c}\~ao pr\'opria.}
315 \end{table}
316
317 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer conteúdo.
318
319 \begin{table}[ht]
320 \centering
321 \caption{Exemplo de Tabela Redimensionada}
322 \label{tab:examplerb}
323 \resizebox{\textwidth}{!}{%
324 \begin{tabular}{lllll}
325 \toprule
326 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 & Coluna 5 \\
327 \midrule
328 Dados 1 & Dados 2 & Dados 3 & Dados 4 & Dados 5 \\
329 Dados 6 & Dados 7 & Dados 8 & Dados 9 & Dados 10 \\
330 \bottomrule
331 \end{tabular}%
332 }
333 \fonte{Elabora\c{c}\~ao pr\'opria.}
334 \end{table}
335
336
337 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotatble|.
338
339 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela.
340
341 \begin{sidewaystable}
342 \centering
343 \caption{Sua Legenda Aqui}
344 \label{tab:sua_tabela}
345 \begin{tabular}{llll}
346 \toprule
347 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
348 \midrule
349 Item 1 & Item 2 & Item 3 & Item 4 \\
350 Item 5 & Item 6 & Item 7 & Item 8 \\
351 \bottomrule
352 \end{tabular}
353 \fonte{Elabora\c{c}\~ao pr\'opria.}
354 \end{sidewaystable}
355
356 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos para
357
358 % Exemplo de tabela longa que se estende por várias páginas
359 \begin{longtable}{|c|c|c|}

```

```

360 % primeiro cabeçalho (é o caption)
361 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
362 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
363 \endfirsthead
364 % cabeçalho normal
365 \multicolumn{3}{c}%
366 {\table\thetable{} -- continuação da página anterior} \\
367 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
368 \endhead
369 % pé normal
370 \hline \multicolumn{3}{r}{{Continua na próxima página}} \\ \hline
371 \endfoot
372 \hline
373 % último pé
374 \multicolumn{3}{r}{{Continua na próxima página}} \\
375 \hline \hline
376 \endlastfoot
377
378 % Conteúdo da tabela
379 1 & 2 & 3 \\
380 4 & 5 & 6 \\
381 1 & 2 & 3 \\
382 4 & 5 & 6 \\
383 1 & 2 & 3 \\
384 4 & 5 & 6 \\
385 1 & 2 & 3 \\
386 4 & 5 & 6 \\
387 1 & 2 & 3 \\
388 4 & 5 & 6 \\
389 1 & 2 & 3 \\
390 4 & 5 & 6 \\
391 1 & 2 & 3 \\
392 4 & 5 & 6 \\
393 1 & 2 & 3 \\
394 4 & 5 & 6 \\
395 1 & 2 & 3 \\
396 1 & 2 & 3 \\
397 4 & 5 & 6 \\
398 1 & 2 & 3 \\
399 4 & 5 & 6 \\
400 1 & 2 & 3 \\
401 4 & 5 & 6 \\
402 1 & 2 & 3 \\
403 4 & 5 & 6 \\
404 1 & 2 & 3 \\
405 4 & 5 & 6 \\
406 1 & 2 & 3 \\
407 4 & 5 & 6 \\
408 1 & 2 & 3 \\
409 4 & 5 & 6 \\
410 1 & 2 & 3 \\
411 4 & 5 & 6 \\
412 1 & 2 & 3 \\
413 4 & 5 & 6

```

414 1 & 2 & 3 \\
 415 4 & 5 & 6 \\
 416 1 & 2 & 3 \\
 417 4 & 5 & 6 \\
 418 1 & 2 & 3 \\
 419 4 & 5 & 6 \\1 & 2 & 3 \\
 420 4 & 5 & 6 \\
 421 1 & 2 & 3 \\
 422 4 & 5 & 6 \\
 423 1 & 2 & 3 \\
 424 1 & 2 & 3 \\
 425 4 & 5 & 6 \\
 426 1 & 2 & 3 \\
 427 4 & 5 & 6 \\
 428 1 & 2 & 3 \\
 429 4 & 5 & 6 \\
 430 1 & 2 & 3 \\
 431 4 & 5 & 6 \\
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 461 4 & 5 & 6 \\
 462 1 & 2 & 3 \\
 463 4 & 5 & 6 \\
 464 1 & 2 & 3 \\
 465 4 & 5 & 6 \\
 466 1 & 2 & 3 \\
 467 4 & 5 & 6 \

```

468 1 & 2 & 3 \\
469 4 & 5 & 6 \\
470 1 & 2 & 3 \\
471 4 & 5 & 6 \\
472 1 & 2 & 3 \\
473 4 & 5 & 6 \\
474 1 & 2 & 3 \\
475 4 & 5 & 6 \\
476 1 & 2 & 3 \\
477 4 & 5 & 6 \\
478 1 & 2 & 3 \\
479 4 & 5 & 6 \\
480 1 & 2 & 3 \\
481 4 & 5 & 6 \\
482 1 & 2 & 3 \\
483 4 & 5 & 6 \\
484 1 & 2 & 3 \\
485 4 & 5 & 6 \\
486
487 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
488 \end{longtable}
489 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
490
491 \section{Subfiguras (subcaption)}
492
493 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
494
495 \begin{figure}[ht]
496   \centering
497   \begin{subfigure}[b]{0.42\textwidth}
498     \centering\includegraphics[width=\linewidth]{example-image-a}
499     \caption{Primeira vista}\label{fig:sub-a}
500   \end{subfigure}
501   \hfill
502   \begin{subfigure}[b]{0.42\textwidth}
503     \centering\includegraphics[width=\linewidth]{example-image-b}
504     \caption{Segunda vista}\label{fig:sub-b}
505   \end{subfigure}
506   \caption{Figura com duas subfiguras}
507   \label{fig:subs}
508   \fonte{Elabora\c{c}\~ao pr\'opria.}
509 \end{figure}
510
511 Referencie a figura inteira com \verb|\autoref{fig:subs}| e cada parte com \verb|\autoref{fig:sub-a}|
512
513 \section{Imagens (graphicx)}
514
515 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
516 \begin{itemize}
517   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
518   \item \verb|[angle=90]| --- rotação;
519   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direita, topo});
520 \end{itemize}
521 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{eps}

```

```

522
523 \section{Introdução ao tikz}
524
525 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
526
527 \begin{figure}[ht]
528   \centering
529   \begin{tikzpicture}
530     \node[draw, rounded corners] (a) at (0,0) {Início};
531     \node[draw, rounded corners] (b) at (4,0) {Fim};
532     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
533   \end{tikzpicture}
534   \caption{Diagrama simples em tikz}
535   \label{fig:tikz}
536   \fonte{Elabora\c{c}\~ao pr\'opria.}
537 \end{figure}
538
539 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|tikz|
540
541 \section{Float ou direto no texto?}
542
543 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \L
544
545 \chapter{Using BibLaTeX}
546 \label{cap:biblatex}
547
548 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
549
550 \textbf{É obrigatório compilar com \texttt{biber}} (e não com \hologo{BibTeX}). A sequênci
551
552 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
553
554 \begin{table}[h]
555   \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
556     \texttt{\textbackslash cite} do \LaTeX\ e
557     o comando \texttt{\textbackslash citet},
558     disponível pela opção \texttt{natbib} do BibLaTeX.}
559   \label{tab:citation}
560   \centering
561   {\footnotesize
562   \begin{tabular}{|c|c|c|}
563     \hline
564     Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
565     \hline
566     Livro & \cite{book-example} & \citet{book-example}\\
567     Artigo & \cite{article-example} & \citet{article-example}\\
568     Relatório & \cite{techreport-example} & \citet{techreport-example}\\
569     Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
570     Anais de Congresso & \cite{inproceedings-example} &
571       \citet{inproceedings-example}\\
572     Séries & \cite{incollection-example} & \citet{incollection-example}\\
573     Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
574     Dissertação de mestrado & \cite{mastersthesis-example} &
575       \citet{mastersthesis-example}

```

```

576 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
577 \hline
578 \end{tabular}}
579 \fonte{Elabora\c{c}\~ao pr\'opria.}
580 \end{table}
581
582 \begin{table}[h]
583 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
584 \texttt{\textbackslash cite} do \LaTeX\ e
585 o comando \texttt{\textbackslash citet},
586 disponivel pela op\c{c}\~ao \texttt{\natbib} do BibLaTeX. Al\~em disso, usando o booktabs.}
587 \label{tab:citation1}
588 \centering
589 {\footnotesize
590 \begin{tabular}{ccc}
591 \toprule
592 Tipo da Publica\c{c}\~ao & \verb|\cite| & \verb|\citet|\\
593 \midrule
594 Livro & \cite{book-example} & \citet{book-example}\\
595 Artigo & \cite{article-example} & \citet{article-example}\\
596 Relat\~orio & \cite{techreport-example} & \citet{techreport-example}\\
597 Relat\~orio & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
598 Anais de Congresso & \cite{inproceedings-example} &
599 \citet{inproceedings-example}\\
600 S\~eries & \cite{incollection-example} & \citet{incollection-example}\\
601 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
602 Disserta\c{c}\~ao de mestrado & \cite{mastersthesis-example} &
603 \citet{mastersthesis-example}\\
604 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
605 \bottomrule
606 \end{tabular}}
607 \fonte{Elabora\c{c}\~ao pr\'opria.}
608 \end{table}
609
610 \section{Comandos de cita\c{c}\~ao}
611
612 Os principais comandos s\~ao: \verb|\cite| (refer\~encia simples), \verb|\citet| (autor por ex
613
614 \subsection{Par\~ametro opcional: p\~agina e ‘‘tradu\c{c}\~ao nossa’’}
615
616 Todos os comandos aceitam um par\~ametro opcional para a localiza\c{c}\~ao (p\~agina, se\c{c}\~ao) e obser
617
618 \subsection{Cita\c{c}\~ao de cita\c{c}\~ao (\emph{apud})}
619
620 Para citar uma obra atrav\~es de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
621
622 \subsection{M\~ultiplas bibliografias}
623
624 Para separar as refer\~encias (por tipo, cap\~itulo ou tema), use o ambiente \verb|refsection|
625
626 \section{Tipos de entrada bibliogr\~afica}
627
628 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sin\~oni
629

```

```

630 \subsection{Tipos acadêmicos principais}
631 \begin{itemize}
632   \item Livro / monografia --- \verb|@book| (\verb|@livro|)
633   \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \verb|@journal|
634   \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
635   \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento --- \verb|@inproceedings|
636   \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo --- \verb|@phdthesis|
637   \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periódico --- \verb|@article|
638   \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/site --- \verb|@misc|
639 \end{itemize}
640
641 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
642 \begin{itemize}
643   \item Patente --- \verb|@patent| (\verb|@patente|)
644   \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato|)
645   \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\verb|@partitura|)
646   \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
647   \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb|@filme|)
648 \end{itemize}
649
650 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
651 \begin{itemize}
652   \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@jogo|)
653   \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetelevisao|, \verb|@serietv|)
654 \end{itemize}
655
656 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|, \verb|assunto|, \verb|palavras-chave|
657
658 \chapter{Alguns outros exemplo úteis}
659
660 \begin{tcolorbox}[title=Meu Textbox]
661 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir imagens, tabelas, etc.
662 \end{tcolorbox}
663
664 \begin{tcolorbox}
665 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como incluir imagens, tabelas, etc.
666 \end{tcolorbox}
667
668 \begin{figure}[ht]
669   \centering
670   \begin{tikzpicture}
671     \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{image.png}}
672     \begin{scope}[x={(image.south east)},y={(image.north west)}]
673       % Definindo o textbox dentro da figura
674       \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=Este textbox fala sobre como inserir um textbox dentro de uma figura usando tikz]
675       \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de Exemplo]
676         Este textbox fala sobre como inserir um textbox dentro de uma figura usando tikz
677       \end{tcolorbox}
678     \end{scope}
679   \end{tikzpicture}
680   \caption{Figura com Textbox}
681   \label{fig:figura_com_textbox1}
682   \fonte{Elabora\c{c}\~ao pr\'opria.}
683

```

```

684 \end{figure}
685
686
687 \begin{figure}[ht]
688     \centering
689     \begin{tcolorbox}
690 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
691 \end{tcolorbox}
692     \caption{Figura com Textbox simples}
693     \label{fig:figura_com_textbox}
694     \fonte{Elabora\c{c}\~ao pr\`opria.}
695 \end{figure}
696
697 \section{Listagens de código e algoritmos}
698
699 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
700
701 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
702 def saudacao(nome):
703     # imprime uma saudação acentuada
704     print(f"Olá, {nome}!")
705
706 saudacao("COPPE")
707 \end{python}
708 \source{Elaboração própria.}
709
710 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
711
712 \begin{gxoutput}
713 Olá, COPPE!
714 \end{gxoutput}
715
716 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando
717
718 \begin{algorithm}[H]
719 \caption{Soma dos elementos de um vetor}
720 \label{alg:soma}
721 \Dados{vetor  $v$  com  $n$  elementos}
722 \Resultado{soma  $s$  dos elementos de  $v$ }
723  $s \leftarrow 0$ ;
724 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
725      $s \leftarrow s + v[i]$ ;
726 }
727 \Retorna  $s$ ;
728 \end{algorithm}
729
730 \chapter{Escrevendo matemática}
731
732 A classe carrega \texttt{amsmath} e \texttt{amssymb}. Fórmulas no texto vão entre \verb|$.
733
734 \section{Alinhamento (align)}
735 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
736 \begin{align}
737     f(x) &= (x+1)^2 \setminus

```

```

738      &= x^2 + 2x + 1 .
739 \end{align}
740
741 \section{Matrizes (matrix)}
742 0 \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
743 \begin{equation}
744   A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
745   \quad
746   \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
747 \end{equation}
748
749 \section{Subequações (subequations)}
750 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
751 \begin{subequations}
752 \begin{align}
753   a &= b + c , \\
754   d &= e - f .
755 \end{align}
756 \end{subequations}
757
758 \section{Estruturas alinhadas (array)}
759 0 \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
760 \begin{equation}
761   |x| = \left\{ \begin{array}{l}
762     x, \quad \text{se } x \geq 0, \\
763     -x, \quad \text{caso contrário.}
764   \end{array} \right.
765 \end{equation}
766
767 \section{Operadores (\textbackslash DeclareMathOperator)}
768 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
769 \begin{verbatim}
770 \usepackage{amsmath}
771 \DeclareMathOperator{\RMSE}{RMSE}
772 \DeclareMathOperator{\diag}{diag}
773 \end{verbatim}
774 Assim, \verb|$\RMSE(y,\hat{y})$| produz  $\text{RMSE}(y,\hat{y})$ , e \verb|$\diag(\lambda_1,\dots$|
775
776 \section{Unicode e LuaLaTeX (unicode-math)}
777 Ao migrar para o \hologo{LuaLaTeX}, troque o \texttt{amssymb} pelo \texttt{unicode-math} e
778 \begin{verbatim}
779 \usepackage{amsmath}
780 \usepackage{unicode-math}
781 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
782 \end{verbatim}
783 0 código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
784
785 \chapter{Referências cruzadas, notas e índices}
786 \label{cap:refs}
787
788 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
789
790 \section{Rótulos e referências cruzadas}
791

```

```

792 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
793
794 \section{Notas de rodapé}
795
796 Use \verb|\footnote{texto}|\footnote{Como esta nota, que a classe imprime em espaçamento s
797
798 \section{hyperref e marcadores}
799
800 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref
801
802 \section{URLs}
803
804 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
805
806 \section{Índice remissivo}
807
808 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
809
810 \section{Traduções (NBR 10520)}
811
812 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
813
814 \chapter{Truques de \LaTeX}
815
816 \section{Comentários mágicos (\texttt{\% !TeX})}
817 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
818 \begin{verbatim}
819 % !TeX program = lualatex
820 % !TeX encoding = UTF-8
821 % !TeX spellcheck = pt_BR
822 % !BIB program = biber
823 \end{verbatim}
824 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
825
826 \section{Espaço engolido}
827 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
828
829 \section{Escopo (grupos)}
830 Chaves \verb|\{...}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
831
832 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash n
833 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
834
835 \section{\texttt{latexmkrc}, perl e Overleaf}
836 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
837
838 \chapter{Método Proposto}
839 \chapter{Resultados e Discuss~oes}
840
841 \section{Algumas Demonstra{\c c}~oes}
842
843 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha
844 \syml[beta]{Beta}{A palavra Beta more e corrigida}
845 \syml[zzbeta]{$\beta$}{A letra $\beta$ corrigida}

```

```

846 \syml{beta}{A palavra beta}
847 \syml{alpha}{A palavra alpha}
848 \syml[alpha]{Alpha}{A palavra Alpha}
849 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
850 \syml[marco]{Marco}{A palavra Marco corrigida}
851
852 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
853 \abbrev{GoT}{Game of Thrones}
854 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
855 \abbrev[iot]{IoT}{IoT ordenado como iot}
856 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
857 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
858 \abbrev{IoT}{IoT com ordenação default}
859 \abbrev[ITU]{ITU}{ITU mesmo}
860
861 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
862 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
863
864 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas automaticamente
865
866
867
868
869 \chapter{Conclusões}
870
871 \backmatter
872
873 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
874 % partitura --- declarados via sinônimos em português em example.bib.
875 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
876 \nocite{jurisprudencia-example}
877 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
878 \nocite{jornalcaderno-example,jornal-example}
879 \printbibliography
880
881
882
883 \appendix
884
885 \chapter{Um apêndice}
886
887 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
888
889 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
890
891
892 \chapter{Outro apêndice}
893
894 \annex
895
896
897 \chapter{Um Anexo}
898 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
899

```

```

900
901
902 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
903
904 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
905
906
907 \chapter{Outro Anexo}
908
909
910 \end{document}
911
912 \example

```

8 Implementation

8.1 The ‘coppe.cls’ file

```

913 \class
914 \def\filename{coppe.dtx}
915 \def\fileversion{v3.8}
916 \def\filedate{2026/05/25}
917 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
918 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
919 \LoadClass[12pt,a4paper,twoside]{book}
920 % The bibliography engine (biblatex+biber) is loaded further below, after babel,
921 % so the numbers class option can select the citation style at load time.
922 \RequirePackage{hyphenat}
923 \RequirePackage{lastpage}
924 \RequirePackage{ifthen}
925 \RequirePackage{graphicx}
926 \RequirePackage{setspace}
927 \RequirePackage{tabularx}
928 \RequirePackage{etoolbox}
929 \RequirePackage{eqparbox}
930 \RequirePackage{ltxcmds}
931 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
932 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
933 \def\TeX{\hologo{TeX}}
934 \def\LaTeX{\hologo{LaTeX}}
935 \def\LaTeXe{\hologo{LaTeXe}}
936 \def\BibTeX{\hologo{BibTeX}}
937 \def\pdfLaTeX{\hologo{pdfLaTeX}}
938 \def\pdfTeX{\hologo{pdfTeX}}
939 \def\LuaLaTeX{\hologo{LuaLaTeX}}
940 \def\XeLaTeX{\hologo{XeLaTeX}}
941 \def\MiKTeX{\hologo{MiKTeX}}
942 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
943 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
944 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
945 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
946 \RequirePackage{iftex}
947 \ifPDFTeX
948 \RequirePackage[utf8]{inputenc}

```

```

949 \AtEndOfClass{\ClassWarningNoLine{coppe}{%
950   Although slower, you should try compiling with LuaLaTeX}}
951 \fi
952 \RequirePackage[T1]{fontenc}
953 % manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
954 % two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.
955 \RequirePackage[a4paper,twoside,bindingoffset=1cm,%
956 top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
957 \RequirePackage{titlesec}
958 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
959 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
960 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
961 % keep "N Title". \appendix and \annex raise the flag (see below).
962 \newif\if@coppeappendix \@coppeappendixfalse
963 \titleformat{\chapter}[hang]
964   {\normalfont\Large\bfseries}
965   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
966   {1ex}}{\MakeUppercase}
967 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
968 \let\coppe@orig@appendix\appendix
969 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
970 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
971 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
972 % which macro is written (so it is correct at .toc-render time even though the
973 % appendixname has changed back); the chapter letter is baked in at write time.
974 \newif\if@coppeannex \@coppeannexfalse
975 \newcommand\coppe@tocapp[1]{%
976   \MakeUppercase{\iflanguage{brazilian}{Ap\^endice}{Appendix}}}%
977   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
978 \newcommand\coppe@tocanx[1]{%
979   \MakeUppercase{\iflanguage{brazilian}{Anexo}{Annex}}}%
980   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
981 \def\@chapter[#1]#2{%
982   \ifnum \c@secnumdepth >\m@ne
983     \if@mainmatter
984       \refstepcounter{chapter}%
985       \typeout{\@chapapp\space\thechapter.}%
986       \if@coppeappendix
987         \if@coppeannex
988           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
989         \else
990           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
991         \fi
992       \else
993         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
994       \fi
995     \else
996       \addcontentsline{toc}{chapter}{#1}%
997     \fi
998   \else
999     \addcontentsline{toc}{chapter}{#1}%
1000 \fi
1001 \chaptermark{#1}%
1002 \addtocontents{lof}{\protect\addvspace{10\p@}}%

```

```

1003 \addtocontents{lot}{\protect\addvspace{10\p@}}%
1004 \if@twocolumn
1005 \topnewpage[\@makechapterhead{#2}]%
1006 \else
1007 \makechapterhead{#2}%
1008 \afterheading
1009 \fi}
1010 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1011 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1012 \titleformat{name=chapter,numberless}[hang]
1013 {\normalfont\Large\bfseries\filcenter}{0pt}{\MakeUppercase}
1014 \titlespacing{name=chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1015 % section: ALL CAPS, normal weight.
1016 \titleformat{section}
1017 {\normalfont\large\bfseries}{\thesection}{1ex}{\MakeUppercase}
1018 % subsection: bold (author writes Title Case).
1019 \titleformat{subsection}
1020 {\normalfont\bfseries}{\thesubsection}{1ex}{}
1021 % subsubsection: bold italic.
1022 \titleformat{subsubsection}
1023 {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
1024 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1025 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1026 \RequirePackage{caption}
1027 \captionsetup{font=footnotesize,labelsep=endash}
1028 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1029 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1030 % at the bottom is mandatory). The float package's plaintop style forces the
1031 % caption above the body for figure, table and the new quadro float. -----
1032 \RequirePackage{float}
1033 \floatstyle{plaintop}
1034 \restylefloat{figure}
1035 \restylefloat{table}
1036 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1037 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1038 % unlike a tabela (rules on top and bottom only).
1039 \providecommand{\quadroname}{\iflanguage{brazilian}{Quadro}{Frame}}
1040 \providecommand{\listquadroname}{%
1041 \iflanguage{brazilian}{Lista de Quadros}{List of Frames}}
1042 \newfloat{quadro}{htbp}{loq}[chapter]
1043 \floatname{quadro}{\quadroname}
1044 \providecommand{\quadroautorefname}{\quadroname}
1045 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1046 % (float's own \l@quadro lays the number/title out incorrectly).
1047 \let\l@quadro\l@figure
1048 \newcommand\listofquadros{%
1049 \chapter*{\listquadroname}%
1050 \addcontentsline{toc}{chapter}{\listquadroname}%
1051 \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1052 \@starttoc{loq}}
1053 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1054 % float as a smaller, single-spaced, centred line; it does NOT advance any
1055 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1056 % the label is "Fonte" (pt) / "Source" (en).

```

```

1057 \newcommand{\cpsourcename}{\iflanguage{brazilian}{Fonte}{Source}}
1058 \newcommand{\cpsource}[1]{%
1059   \par\addvspace{\smallskipamount}%
1060   {\footnotesize\singlespacing\centering\cpsourcename: #1\par}}
1061 \let\cpfonte\cpsource
1062 \@ifundefined{source}{\let\source\cpsource}{}%
1063 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1064 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1065 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1066 \newcommand{\newcoppefloat}[3]{%
1067   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1068   \floatname{#1}{#2}%
1069   \expandafter\newcommand\csname listof#1s\endcsname{%
1070     \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
1071     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1072     \@starttoc{lo#1}}}%
1073 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1074 %   (filete) from the left margin; footnotesize is the book default. -----
1075 \let\coppe@orig@footnotetext@footnotetext
1076 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1077 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1078 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1079 %   (manual2024 2.6 / proposta P24). -----
1080 \RequirePackage{enumitem}
1081 \newlist{alineas}{enumerate}{2}
1082 \setlist[alineas,1]{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1083 \setlist[alineas,2]{label=--,leftmargin=1.8em,itemsep=0pt}
1084 % --- Page numbering: arabic in the top OUTER corner on textual pages
1085 %   (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1086 %   \frontmatter (unnumbered by default; roman with the numeraisromanos
1087 %   class option). -----
1088 \RequirePackage{fancyhdr}
1089 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}
1090 \renewcommand{\headrulewidth}{0pt}
1091 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1092 % number may leak onto them (e.g. the verso right after the cover).
1093 \renewcommand{\cleardoublepage}{%
1094   \clearpage
1095   \if@twoside
1096     \ifodd\c@page\else
1097       \hbox{}\thispagestyle{empty}\newpage
1098     \if@twocolumn\hbox{}\newpage\fi
1099   \fi
1100   \fi}
1101 \def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1102 p\kern-.025em e}}\kern-.08em
1103 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
1104 \newboolean{maledoc}
1105 \setboolean{maledoc}{false}
1106 %
1107 % Class options.
1108 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1109 % fixed language slots:
1110 % * main      -- the language of the body, captions, and the main abstract.

```

```

1111 %           Set by a class option: brazilian (default), english,
1112 %           spanish, french, italian, or any user-supplied language
1113 %           that ships a coppe-lang-<name>.def file.
1114 % * foreign -- the language used by the foreignabstract environment and
1115 %           by \foreign@... registers. Defaults to english (or to
1116 %           brazilian when main=english) for back-compat.
1117 % * third   -- optional, loaded on demand with \usecoppelanguage{<name>}
1118 %           in the preamble (for citations or quotations in a third
1119 %           language). Has no class option of its own.
1120 % Backward compatibility: the historical \if@english boolean is kept and
1121 % mirrors "is main = english?", so every existing \if@english branch in the
1122 % class keeps working unchanged.
1123 \def\coppe@mainlang{brazilian}
1124 \def\coppe@foreignlang{english}
1125 \newif\if@english\@englishfalse
1126 \DeclareOption{brazilian}{%
1127   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}\@englishfalse}
1128 \DeclareOption{english}{%
1129   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}\@englishtrue}
1130 \DeclareOption{spanish}{%
1131   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}\@englishfalse}
1132 \DeclareOption{french}{%
1133   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}\@englishfalse}
1134 \DeclareOption{italian}{%
1135   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}\@englishfalse}
1136 \DeclareOption{msc}{%
1137   \newcommand{\@degree}{M.Sc.}
1138   \newcommand{\@degreename}{Mestrado}
1139   \newcommand{\local@degname}{Mestre}
1140   \newcommand{\foreign@degname}{Master}
1141   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1142   \newcommand\foreign@doctype{Dissertation}
1143 }
1144 \DeclareOption{dsccexam}{%
1145   \newcommand{\@degree}{D.Sc.}
1146   \newcommand{\@degreename}{Doutorado}
1147   \newcommand{\local@degname}{Doutor}
1148   \newcommand{\foreign@degname}{Doctor}
1149   \setboolean{maledoc}{true}
1150   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1151   \newcommand\foreign@doctype{Qualifying Exam}
1152 }
1153 \DeclareOption{mscexam}{%
1154   \newcommand{\@degree}{M.Sc.}
1155   \newcommand{\@degreename}{Mestrado}
1156   \newcommand{\local@degname}{Mestre}
1157   \newcommand{\foreign@degname}{Master}
1158   \setboolean{maledoc}{true}
1159   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1160   \newcommand\foreign@doctype{Qualifying Exam}
1161 }
1162 \DeclareOption{dsc}{%
1163   \newcommand{\@degree}{D.Sc.}
1164   \newcommand{\@degreename}{Doutorado}

```

```

1165 \newcommand{\local@degname}{Doutor}
1166 \newcommand{\foreign@degname}{Doctor}
1167 \newcommand\local@doctype{Tese}
1168 \newcommand\foreign@doctype{Thesis}
1169 }
1170 \newif\if@coppenumeric\@coppenumericfalse
1171 \DeclareOption{numbers}{\@coppenumerictrue}
1172 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1173 % pre-textual pages are counted but not numbered (manual2024 2.7).
1174 \newif\if@copperoman\@copperomanfalse
1175 \DeclareOption{numeraisromanos}{\@copperomantrue}

Here is the default one-and-a-half line spacing. Users can change to double spacing
by passing the doublespacing option.
1176 \onehalfspacing
1177 \DeclareOption{doublespacing}{%
1178   \doublespacing
1179 }

1180 \ProcessOptions\relax
1181 % Built-in language names (used by the loader and the dispatcher below).
1182 \def\coppe@@brazilianstr{brazilian}
1183 \def\coppe@@englishstr{english}
1184 % Load Babel with the foreign language first and the main language last
1185 % (Babel treats the LAST option in the list as the main language). With
1186 % pt-main this produces [english,brazilian]{babel} and with en-main
1187 % [brazilian,english] -- identical to the historical \if@english calls.
1188 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1189 \expandafter\RequirePackage\expandafter[\coppe@babelopts]{babel}
1190 % Auto-load language packs for any non-built-in language listed in babelopts.
1191 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1192 % and need no extra file; others (spanish/french/italian/...) ship as
1193 % coppe-lang-<name>.def alongside the class.
1194 \for\coppe@@lang:=\coppe@babelopts\do{%
1195   \edef\coppe@@cur{\coppe@@lang}%
1196   \ifx\coppe@@cur\coppe@@brazilianstr\else
1197     \ifx\coppe@@cur\coppe@@englishstr\else
1198       \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1199       {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1200       {\ClassError{coppe}
1201         {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1202         {Install it next to coppe.cls, or pick a built-in language.}}%
1203     \fi\fi}
1204 % --- Multilingual string dispatcher (scaffolding -- not yet wired in) -----
1205 \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1206 % language <lang>. The three retrievers below look strings up either at the
1207 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1208 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1209 \providecommand{\copperdefstring}[3]{%
1210   \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1211 \providecommand{\coppestring}[1]{\@nameuse{coppe@str@\language@ #1}}
1212 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@\coppe@mainlang @#1}}
1213 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@\coppe@foreignlang @#1}}
1214 % Load an extra language pack at any point in the preamble (third-language
1215 % slot for citations or quotations beyond the main+foreign pair).

```

```

1216 \providecommand{\usecoppelanguage}[1]{%
1217   \def\coppe@@cur{#1}%
1218   \ifx\coppe@@cur\coppe@@brazilianstr\else
1219   \ifx\coppe@@cur\coppe@@englishstr\else
1220     \InputIfFileExists{coppe-lang-#1.def}{}
1221     {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1222   \fi\fi}
1223 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1224 % consulted by the cover/folha renderers once they are migrated in step 2.
1225 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1226 % still drive every visible site, so behavior for pt/en is unchanged.
1227 \providecommand{\titlein}[2]{%
1228   \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1229 % --- Built-in string tables: brazilian + english. The values mirror the
1230 % literals currently scattered through the class; step-1 only populates the
1231 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1232 \copperdefstring{brazilian}{appendix}{Ap\~endice}
1233 \copperdefstring{english}{appendix}{Appendix}
1234 \copperdefstring{brazilian}{annex}{Anexo}
1235 \copperdefstring{english}{annex}{Annex}
1236 \copperdefstring{brazilian}{frame}{Quadro}
1237 \copperdefstring{english}{frame}{Frame}
1238 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1239 \copperdefstring{english}{listframe}{List of Frames}
1240 \copperdefstring{brazilian}{source}{Fonte}
1241 \copperdefstring{english}{source}{Source}
1242 \copperdefstring{brazilian}{program}{Programa}
1243 \copperdefstring{english}{program}{Program}
1244 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1245 \copperdefstring{english}{listprogram}{List of Programs}
1246 \copperdefstring{brazilian}{references}{Refer\~encias}
1247 \copperdefstring{english}{references}{References}
1248 \copperdefstring{brazilian}{in:}{em}
1249 \copperdefstring{english}{in:}{in}
1250 \copperdefstring{brazilian}{art}{Ilustra\c{c}\~ao}
1251 \copperdefstring{english}{art}{Art}
1252 \copperdefstring{brazilian}{platform}{Plataforma}
1253 \copperdefstring{english}{platform}{Platform}
1254 \copperdefstring{brazilian}{directedby}{Dire\c{c}\~ao}
1255 \copperdefstring{english}{directedby}{Directed by}
1256 \copperdefstring{brazilian}{producedby}{Produ\c{c}\~ao}
1257 \copperdefstring{english}{producedby}{Produced by}
1258 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1259 \copperdefstring{english}{listabbreviation}{List of Abbreviations and Acronyms}
1260 \copperdefstring{brazilian}{listsymbol}{Lista de S{\~i}mbolos}
1261 \copperdefstring{english}{listsymbol}{List of Symbols}
1262 \copperdefstring{brazilian}{glossary}{Gloss{\~a}rio}
1263 \copperdefstring{english}{glossary}{Glossary}
1264 \copperdefstring{brazilian}{advisor}{Orientador}
1265 \copperdefstring{english}{advisor}{Advisor}
1266 \copperdefstring{brazilian}{advisors}{Orientadores}
1267 \copperdefstring{english}{advisors}{Advisors}
1268 \copperdefstring{brazilian}{dept}{Programa}
1269 \copperdefstring{english}{dept}{Department}

```

```

1270 \copperdefstring{brazilian}{approvedmale} {Examinado por}
1271 \copperdefstring{brazilian}{approvedfemale}{Examinada por}
1272 \copperdefstring{english} {approvedmale} {Examined by}
1273 \copperdefstring{english} {approvedfemale}{Examined by}
1274 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1275 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1276 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}
1277 \copperdefstring{brazilian}{cityname} {Rio de Janeiro}
1278 \copperdefstring{english} {cityname} {Rio de Janeiro}
1279 \copperdefstring{brazilian}{statename} {RJ}
1280 \copperdefstring{english} {statename} {RJ}
1281 \copperdefstring{brazilian}{countryname} {Brasil}
1282 \copperdefstring{english} {countryname} {Brasil}
1283 % Month names.
1284 \copperdefstring{brazilian}{monthname1} {Janeiro}
1285 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1286 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1287 \copperdefstring{brazilian}{monthname4} {Abril}
1288 \copperdefstring{brazilian}{monthname5} {Maio}
1289 \copperdefstring{brazilian}{monthname6} {Junho}
1290 \copperdefstring{brazilian}{monthname7} {Julho}
1291 \copperdefstring{brazilian}{monthname8} {Agosto}
1292 \copperdefstring{brazilian}{monthname9} {Setembro}
1293 \copperdefstring{brazilian}{monthname10}{Outubro}
1294 \copperdefstring{brazilian}{monthname11}{Novembro}
1295 \copperdefstring{brazilian}{monthname12}{Dezembro}
1296 \copperdefstring{english}{monthname1} {January}
1297 \copperdefstring{english}{monthname2} {February}
1298 \copperdefstring{english}{monthname3} {March}
1299 \copperdefstring{english}{monthname4} {April}
1300 \copperdefstring{english}{monthname5} {May}
1301 \copperdefstring{english}{monthname6} {June}
1302 \copperdefstring{english}{monthname7} {July}
1303 \copperdefstring{english}{monthname8} {August}
1304 \copperdefstring{english}{monthname9} {September}
1305 \copperdefstring{english}{monthname10}{October}
1306 \copperdefstring{english}{monthname11}{November}
1307 \copperdefstring{english}{monthname12}{December}
1308 % algorithm2e package option for this language (empty -> english default).
1309 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1310 \copperdefstring{english} {algorithm2eopt}{}
1311 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1312 \copperdefstring{brazilian}{babelname}{brazilian}
1313 \copperdefstring{english} {babelname}{english}
1314 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1315 % Portuguese alias for short, inline quotations (proposta P3).
1316 \RequirePackage{csquotes}
1317 \newcommand{\citacao}[1]{\enquote{#1}}
1318 % --- Code listings & algorithms (TODO3 "tratar listagens"): the listings
1319 % package gives the "Programa" listing (caption on top, ABNT R46) with
1320 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1321 % the language styles below are based on the user's listagens.tex. -----
1322 \RequirePackage{xcolor}
1323 \RequirePackage{listings}

```

```

1324 \RequirePackage{fancyvrb}
1325 \definecolor{deepblue}{rgb}{0,0,0.5}
1326 \definecolor{deepred}{rgb}{0.6,0,0}
1327 \definecolor{deepgreen}{rgb}{0,0.5,0}
1328 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
1329 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
1330 \newcommand{\gxnumbersep}{0.5em}
1331 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1332   literate=%
1333   {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
1334   {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
1335   {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
1336   {\~}{\~{C}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
1337   {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{O}}1%
1338   {\~}{\~{U}}1%
1339 }
1340 % Babel-aware listing names, set after listings' own babel setup.
1341 \AtBeginDocument{%
1342   \renewcommand\lstlistingname{\iflanguage{brazilian}{Programa}{Program}}%
1343   \renewcommand\lstlistlistingname{%
1344     \iflanguage{brazilian}{Lista de Programas}{List of Programs}}%
1345 % \listofprogramas in the coppe list style (heading + ToC + running head).
1346 \newcommand\listofprogramas{%
1347   \chapter*{\lstlistlistingname}%
1348   \addcontentsline{toc}{chapter}{\lstlistlistingname}%
1349   \mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1350   \starttoc{lol}}
1351 % Per-language styles + environments (python, java, xml, html, prolog).
1352 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1353   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
1354   morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
1355   emph={MyClass,__init__},emphstyle=\ttb\color{deepred},%
1356   stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
1357   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1358   escapechar=',postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1359 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}]%
1360   \csname\@lst @SetFirstNumber\endcsname{\csname\@lst @SaveFirstNumber\endcsname}
1361   \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!]
1362 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}]
1363 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1364   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
1365   morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1366   keywordstyle=\ttb\color{deepblue},%
1367   emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1368   emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
1369   showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1370   postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1371 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{}
1372 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1373   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
1374   keywordstyle=\ttb\color{deepblue},emphstyle=\ttb\color{deepred},%
1375   stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
1376   extendedchars=true,inputencoding=utf8,breaklines=true,%
1377   postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}

```

```

1378 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{%
1379 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1380   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
1381   keywordstyle=\ttb\color{deepblue},emph={MyClass, __init__},%
1382   emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
1383   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1384   breaklines=true,escapechar=',%
1385   postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}%
1386 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
1387   \csname\@lst @SetFirstNumber\endcsname\{\csname\@lst @SaveFirstNumber\endcsname\}
1388 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1389   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
1390   morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
1391   emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
1392   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1393   breaklines=true,escapechar=',%
1394   postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}%
1395 \lstnewenvironment{java}[1][\javastyle\lstset{#1}%
1396   \csname\@lst @SetFirstNumber\endcsname\{\csname\@lst @SaveFirstNumber\endcsname\}
1397 \newcommand\javainline[1]{\javastyle\lstinline!#1!}%
1398 \newcommand\javaexternal[2][\javastyle\lstinputlisting{#1}{#2}}%
1399 % Verbatim output environments (program output), fancyvrb.
1400 \DefineVerbatimEnvironment{gxoutput}{Verbatim}%
1401   {frame=lines,numbers=left,numbersep=\gxnumbersep}%
1402 \DefineVerbatimEnvironment{gxoutputs}{Verbatim}%
1403   {frame=lines,numbers=left,numbersep=\gxnumbersep,fontsize=\small}%
1404 \renewcommand\theFancyVerbLine{\small\ttfamily\arabic{FancyVerbLine}}%
1405 % Algorithms: algorithm2e, language following the english class option.
1406 \if@english
1407   \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1408 \else
1409   \RequirePackage[portuguese,linesnumbered,lined,algochapter,ruled]{algorithm2e}
1410 \fi
1411 \SetKwFor{Para}{para}{faça}{fim do para}
1412 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1413 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
1414 % former natbib+bibtex setup. Loaded here, after babel, so its language strings
1415 % map correctly. The numbers option selects the numeric style; otherwise the
1416 % author-date default is used. natbib=true keeps \citel/\citep available.
1417 \if@coppenumeric
1418   \RequirePackage[backend=biber,datamodel=coppe,%
1419     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
1420 \else
1421   \RequirePackage[backend=biber,datamodel=coppe,%
1422     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
1423 \fi
1424 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
1425 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
1426 % "Bibliografia" (babel's default \bibname). The title is set directly with
1427 % \iflanguage so it is correct in BOTH the chapter heading and the .toc entry:
1428 % \iflanguage resolves to plain text and survives the write to the .toc, whereas
1429 % biblatex's \bibstring does not, and babel's \bibname caption override proved
1430 % unreliable here. Evaluated in the body, the current language is settled.
1431 \defbibheading{bibliography}{%

```

```

1432 \chapter*{\iflanguage{brazilian}{Refer\~encias}{References}}%
1433 \addcontentsline{toc}{chapter}{\iflanguage{brazilian}{Refer\~encias}{References}}
1434 % The catalog page range (bib:begin/bib:end) was written by the old
1435 % thebibliography redefinition, which biblatex's \printbibliography does not
1436 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
1437 % begin label uses that hook and the end label is emitted by wrapping
1438 % \printbibliography to write it just after the reference list.
1439 \AtBeginBibliography{\coppe@bibBegin}
1440 \let\coppe@orig@printbibliography\printbibliography
1441 \renewcommand\printbibliography[1][]{\%
1442 \coppe@orig@printbibliography[#1]\%
1443 \coppe@bibEnd}

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1444 \newcommand\department[1]{\%
1445 \ifthenelse{\equal{#1}{PEB}}{
1446 {\global\def\local@deptname{Engenharia Biom(\` e)dica}
1447 \global\def\foreign@deptname{Biomedical Engineering}}{
1448 \ifthenelse{\equal{#1}{PEC}}{
1449 {\global\def\local@deptname{Engenharia Civil}
1450 \global\def\foreign@deptname{Civil Engineering}}{
1451 \ifthenelse{\equal{#1}{PEE}}{
1452 {\global\def\local@deptname{Engenharia El(\` e)trica}
1453 \global\def\foreign@deptname{Electrical Engineering}}{
1454 \ifthenelse{\equal{#1}{PEM}}{
1455 {\global\def\local@deptname{Engenharia Mec(\` a)nica}
1456 \global\def\foreign@deptname{Mechanical Engineering}}{
1457 \ifthenelse{\equal{#1}{PEMM}}{
1458 {\global\def\local@deptname{Engenharia Metal(\` u)rgica e de Materiais}
1459 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{
1460 \ifthenelse{\equal{#1}{PEN}}{
1461 {\global\def\local@deptname{Engenharia Nuclear}
1462 \global\def\foreign@deptname{Nuclear Engineering}}{
1463 \ifthenelse{\equal{#1}{PEN0}}{
1464 {\global\def\local@deptname{Engenharia Oce(\` a)nica}
1465 \global\def\foreign@deptname{Ocean Engineering}}{
1466 \ifthenelse{\equal{#1}{PPE}}{
1467 {\global\def\local@deptname{Planejamento Energ(\` e)tico}
1468 \global\def\foreign@deptname{Energy Planning}}{
1469 \ifthenelse{\equal{#1}{PEP}}{
1470 {\global\def\local@deptname{Engenharia de Produ(c c){\~ a}o}
1471 \global\def\foreign@deptname{Production Engineering}}{
1472 \ifthenelse{\equal{#1}{PEQ}}{
1473 {\global\def\local@deptname{Engenharia Qu(\` i)mica}
1474 \global\def\foreign@deptname{Chemical Engineering}}{
1475 \ifthenelse{\equal{#1}{PESC}}{
1476 {\global\def\local@deptname{Engenharia de Sistemas e Computa(c c){\~ a}o}
1477 \global\def\foreign@deptname{Systems Engineering and Computer Science}}{
1478 \ifthenelse{\equal{#1}{PET}}{
1479 {\global\def\local@deptname{Engenharia de Transportes}
1480 \global\def\foreign@deptname{Transportation Engineering}}{
1481 \ifthenelse{\equal{#1}{PENT}}{

```

```

1482     {\global\def\local@deptname{Engenharia de Nanotecnologia}
1483      \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1484 }

\title Used to enter the title in Brazilian Portuguese.
1485 \renewcommand\title[1]{%
1486   \global\def\local@title{#1}%
1487 }

\foreigntitle Used to enter the foreign title.
1488 \newcommand\foreigntitle[1]{%
1489   \global\def\foreign@title{#1}%
1490 }

\advisor Defines globally the title, name and academic degree of the advisors.
1491 \newcount\@advisor\@advisor0
1492 \newcommand\advisor[4]{%
1493   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
1494   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
1495   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
1496   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
1497   \global\advance\@advisor by 1
1498   \ifnum\@advisor>1
1499     \renewcommand\local@advisorstring{Orientadores}
1500     \renewcommand\foreign@advisorstring{Advisors}
1501   \fi
1502 }

\examiner
1503 \newcount\@examiner\@examiner0
1504 \newcommand\examiner[3]{%
1505   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
1506   \global\advance\@examiner by 1
1507 }

\author It was redefined to allow the identification of the author's first names and surname.
1508 \renewcommand\author[2]{%
1509   \global\def\@authname{#1}
1510   \global\def\@authsurn{#2}
1511 }

\date This code makes easy to switch from dates in different languages.
1512 \renewcommand\date[2]{%
1513   \month=#1
1514   \year=#2
1515 }

\local@monthname
1516 \newcommand\local@monthname{\ifcase\month\or
1517   Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
1518   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}

```

\foreign@monthname

```
1519 \newcommand\foreign@monthname{\ifcase\month\or
1520   January\or February\or March\or April\or May\or June\or
1521   July\or August\or September\or October\or November\or December\fi}
```

\keyword

```
1522 \newcounter{keywords}
1523 \newcommand\keyword[1]{%
1524   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1525   \global\addtocounter{keywords}{1}
1526 }
```

\freeconfig This command allows easy changing of core class parameters.

```
1527 \newcommand\freeconfig[6]{%
1528   \providecommand\@degree{}
1529   \renewcommand\@degree{#1}
1530   \providecommand\@degreename{}
1531   \renewcommand\@degreename{#2}
1532   \providecommand\local@degname{}
1533   \renewcommand\local@degname{#3}
1534   \providecommand\foreign@degname{}
1535   \renewcommand\foreign@degname{#4}
1536   \providecommand\local@doctype{}
1537   \renewcommand\local@doctype{#5}
1538   \providecommand\foreign@doctype{}
1539   \renewcommand\foreign@doctype{#6}%
1540 }
1541 %   \end{macrocode}
1542 % \end{macro}
1543 %
1544 % \begin{macro}{\frontmatter}
1545 % The number of pages for both frontmatter and mainmatter printed
1546 % in the cataloging details page is computed by means of simple
1547 % \LaTeX\ labels.
1548 %   \begin{macrocode}
1549 \renewcommand\frontmatter{%
1550   \cleardoublepage
1551   \@mainmatterfalse
1552   \pagenumbering{roman}
1553   \thispagestyle{empty}
1554   % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
1555   % start keeps the page-counter parity aligned with the physical (recto) side,
1556   % so \cleardoublepage lands every pre-textual element on an odd (front) page.
1557   \setcounter{page}{1}
1558   \makefrontpage
1559   \clearpage
1560   \if@copperoman
1561     \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1562     \renewcommand{\headrulewidth}{0pt}}%
1563   \pagestyle{coppe}%
1564   \else
1565     \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
1566     \pagestyle{empty}%
1567   \fi
```

```

1568 }

\mainmatter
1569 \renewcommand\mainmatter{%
1570   \coppe@mainBegin
1571   \cleardoublepage
1572   \@mainmattertrue
1573   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
1574     \renewcommand{\headrulewidth}{0pt}}%
1575   \pagestyle{coppe}%
1576   \pagenumbering{arabic}}

\backmatter
1577 \renewcommand\backmatter{%
1578   \if@openright
1579     \cleardoublepage
1580   \else
1581     \clearpage
1582   \fi}
1583 %

\makecover The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is
the institution block — the university, the full COPPE name, and the graduate
program taken from \department — followed by the author, the title, and the
city and year, all centred and uppercased. The optional UFRJ and COPPE logos
head the page. The capa is not counted: it precedes the (roman) front matter and
is produced before the titlepage (folha de rosto).
1584 \newcommand\makecover{%
1585   \begin{titlepage}%
1586   \centering
1587   \makebox[\linewidth]{%
1588     \includegraphics[height=2cm]{ufrj-logo}\hfill
1589     \includegraphics[height=2cm]{coppe-logo}}\par
1590   \vspace{1.5cm}
1591   {\singlespacing
1592     \MakeUppercase{\local@universityname}\par
1593     \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de
1594       P{\~{o}}s-Gradua{\c c}{\~{a}}o e Pesquisa de Engenharia (COPPE)}}\par
1595     \MakeUppercase{Programa de \local@deptname}\par}
1596   \vspace{\stretch{2}}
1597   \textbf{\MakeUppercase{\@authname\ \@authsur}}\par
1598   \vspace{\stretch{2}}
1599   \nohyphens{%
1600     \if@english
1601       \MakeUppercase\foreign@title
1602     \else
1603       \MakeUppercase\local@title
1604     \fi}\par
1605   \vspace{\stretch{3}}
1606   \MakeUppercase{\local@cityname}\par
1607   \number\year
1608   \end{titlepage}%
1609 }

```

\maketitle

```
1610 \renewcommand\maketitle{%
1611   \pagenumbering{alph}
1612   \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
1613   \makecover
1614   \begin{titlepage}
1615   \begin{flushleft}
1616     \vspace*{1.5mm}
1617     \setlength\baselineskip{0pt}
1618     \setlength\parskip{1mm}
1619     \makebox[\linewidth]{%
1620       \includegraphics[height=2cm]{ufrj-logo}\hfill
1621       \includegraphics[height=2cm]{coppe-logo}}
1622     \end{flushleft}
1623     \vspace{1.05cm}
1624     \begin{center}
1625       \nohyphens{%
1626         \if@english
1627           \MakeUppercase\foreign@title
1628         \else
1629           \MakeUppercase\local@title
1630         \fi}\par
1631       \vspace*{3cm}
1632       \nohyphens{\@authname\ \@authsurn}\par
1633     \end{center}
1634     \vspace*{2.1cm}
1635     \begin{flushright}
1636       \begin{minipage}{8.45cm}
1637         \frontcover@maintext
1638       \end{minipage}\par
1639       \vspace*{7.5mm}
1640       \nohyphens{%
1641         \begin{tabularx}{8.45cm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1642           \local@advisorstring: &
1643           \count1=0
1644           \toks@={}
1645           \@whilenum \count1<\@advisor \do{%
1646             \ifcase\count1 % same as \ifnum0=\count1
1647               \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1648                 \expandafter\endcsname\expandafter\space%
1649                 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1650             \else
1651               \toks@=\expandafter\expandafter\expandafter{%
1652                 \expandafter\the\expandafter\toks@%
1653                 \expandafter&\expandafter\space%
1654                 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1655                 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1656             }%
1657             \fi
1658             \advance\count1 by 1}
1659           \the\toks@
1660         \end{tabularx}}\par
1661       \end{flushright}
1662       \vspace*{\fill}
```

```

1663 \begin{center}
1664 \local@cityname\par
1665 \local@monthname\ de \number\year
1666 \end{center}
1667 \end{titlepage}
1668 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
1669 % the title page is a recto, so this lands on the immediately following verso,
1670 % ahead of the approval sheet produced by \frontmatter.
1671 \ifthenelse{\boolean{maledoc}}{\thispagestyle{empty}\makecatalog}%
1672 \global\let\maketitle\relax%
1673 \global\let\and\relax}

1674 \newcommand\makefrontpage{%
1675 \begin{center}
1676 \sloppy\nohyphens{
1677 \if@english
1678 \MakeUppercase\foreign@title
1679 \else
1680 \MakeUppercase\local@title
1681 \fi}\par
1682 \vspace*{7mm}
1683 {\@authname\ \@authsur}\par
1684 \end{center}\par
1685 \vspace*{4mm}
1686 \frontpage@maintext
1687 \vspace*{16mm}
1688 \nohyphens{%
1689 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}
1690 \local@advisorstring: &
1691 \count1=0
1692 \toks@={}
1693 \@whilenum \count1<\@advisor \do{%
1694 \ifcase\count1 % same as \ifnum0=\count1
1695 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1696 \expandafter\endcsname\expandafter\space%
1697 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1698 \else
1699 \toks@=\expandafter\expandafter\expandafter{%
1700 \expandafter\the\expandafter\toks@%
1701 \expandafter&\expandafter\space%
1702 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1703 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1704 }%
1705 \fi
1706 \advance\count1 by 1}
1707 \the\toks@
1708 \end{tabularx}}\par
1709 \vspace*{20mm}
1710 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}
1711 Aprovada por: &
1712 \count1=0
1713 \toks@={}
1714 \@whilenum \count1<\@examiner \do{%
1715 \ifcase\count1 % same as \ifnum0=\count1

```

```

1716     \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
1717     \expandafter\endcsname\expandafter\}}
1718   \else
1719     \toks@=\expandafter\expandafter\expandafter{%
1720     \expandafter\the\expandafter\toks@%
1721     \expandafter&\expandafter\space%
1722     \csname CoppeExaminer:\the\count1\expandafter\endcsname%
1723     \expandafter\space\}
1724   }%
1725   \fi
1726   \advance\count1 by 1}
1727   \the\toks@
1728   \end{tabularx}}\par
1729   \vspace*{\fill}
1730   \frontpage@bottomtext}

1731 \newcommand\coppe@hypersetup{%
1732 \begingroup
1733 % changes to \toks@ and \count@ are kept local;
1734 % it's not necessary for them, but it is usually the case
1735 % for \count1, because the first ten counters are written
1736 % to the DVI file, thus you got lucky because of PDF output
1737 \toks@={}% in this special case not necessary
1738 \count@=0 %
1739 \@whilenum\count@<\value{keywords}\do{%
1740   % * a keyword separator is not necessary,
1741   %   if there is just one keyword
1742   % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
1743   %   at least once, to get rid of the loop depended \count@
1744   \ifcase\count@ % same as \ifnum0=\count@
1745     \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
1746   \else
1747     \toks@=\expandafter\expandafter\expandafter{%
1748     \expandafter\the\expandafter\toks@
1749     \expandafter;\expandafter\space
1750     \csname CoppeKeyword:\the\count@\endcsname
1751     }%
1752   \fi
1753   \advance\count@ by 1 %
1754 }%
1755 \edef\x{\endgroup
1756   \noexpand\hypersetup{%
1757     pdfkeywords={\the\toks@}%
1758   }%
1759 }%
1760 \x
1761 \hypersetup{%
1762   pdfauthor={\@authname\ \@authsur},
1763   pdftitle={\local@title},
1764   pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
1765   pdfcreator={LaTeX with CoppeTeX toolkit},
1766   breaklinks={true},
1767   raiselinks={true},
1768   pageanchor={true},

```

1769 }}

`\makecatalog` When the document has illustrations, it is required to insert “: il.” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\@isundefined` [?].

```

1770 \newcommand\makecatalog{%
1771   \vspace*{\fill}
1772   \begin{center}
1773     \setlength{\fboxsep}{5mm}
1774     \framebox[120mm][c]{\makebox[5mm][c]{}%
1775       \begin{minipage}[c]{105mm}
1776         \setlength{\parindent}{5mm}
1777         \noindent\sloppy\nohyphens\@authsurn,
1778         \nohyphens\@authname\par
1779         \nohyphens{%
1780           \if@english
1781             \foreign@title%
1782           \else
1783             \local@title%
1784             \fi/\@authname\ \@authsurn. -- \local@cityname:
1785         UFRJ/COPPE, \number\year.}\par
1786         \pageref{front:pageno},
1787         \pageref{LastPage}
1788         p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}} $29,7$cm.\par
1789         % There is an issue here. When the last entry must be split between lines,
1790         % the spacing between it and the next paragraph becomes smaller.
1791         % Should we manually introduce a fixed space? But how could we know that
1792         % a name was split? Is this happening yet?
1793         \nohyphens{%
1794           \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1795             \local@advisorstring: &
1796             \count1=0
1797             \toks@={}
1798             \@whilenum \count1<\@advisor \do{%
1799               \ifcase\count1 % same as \ifnum0=\count1
1800                 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1801                   \expandafter\endcsname\expandafter\space%
1802                   \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1803               \else
1804                 \toks@=\expandafter\expandafter\expandafter{%
1805                   \expandafter\the\expandafter\toks@
1806                   \expandafter&\expandafter\space
1807                   \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1808                   \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1809               }%
1810             \fi
1811             \advance\count1 by 1}
1812             \the\toks@
1813           \end{tabularx}}\par
1814           \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1815           UFRJ/COPPE/Programa de \local@deptname, \number\year.}\par
1816           Refer{\^ e}ncias Bibliogr{\^ a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1817           \count1=0

```

```

1818 \count2=1
1819 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1820 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1821 \advance\count1 by 1
1822 \advance\count2 by 1}
1823 I. \csname CoppeAdvisorSurname:0\endcsname,%
1824 \ \csname CoppeAdvisorName:0\endcsname%
1825 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
1826 II. \local@universityname, COPPE, Programa de \local@deptname.
1827 III. T{\ ' i}tulo.}
1828 \end{minipage}}
1829 \end{center}
1830 \vspace*{\fill}}

```

\dedication

```

1831 \newcommand\dedication[1]{
1832 \gdef\@dedic{#1}
1833 \cleardoublepage
1834 \vspace*{\fill}
1835 \begin{flushright}
1836 \begin{minipage}{60mm}
1837 \raggedleft \it \normalsize \@dedic
1838 \end{minipage}
1839 \end{flushright}}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

1840 \newenvironment{abstract}{%
1841 \cleardoublepage
1842 \thispagestyle{plain}
1843 \abstract@toptext\par
1844 \vspace*{8.6mm}
1845 \begin{center}
1846 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1847 \vspace*{13.2mm}
1848 \@authname\ \@authsurn \par
1849 \vspace*{7mm}
1850 \local@monthname/\number\year
1851 \end{center}\par
1852 \vspace*{1.5cm}
1853 \noindent%
1854 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}
1855 \local@advisorstring: &
1856 \count1=0
1857 \toks@={}
1858 \@whilenum \count1<\@advisor \do{%
1859 \ifcase\count1 % same as \ifnum0=\count1
1860 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1861 \expandafter\endcsname\expandafter\space%
1862 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1863 \else
1864 \toks@=\expandafter\expandafter\expandafter{%
1865 \expandafter\the\expandafter\toks@
1866 \expandafter&\expandafter\space
1867 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%

```

```

1868      \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1869      }%
1870      \fi
1871      \advance\count1 by 1}
1872      \the\toks@
1873 \end{tabularx}\par
1874 \vspace*{2mm}
1875 \noindent\local@deptstring: \local@deptname\par
1876 \vspace*{7mm}}{\vspace*{\fill}}

```

foreignabstract (*env.*)

```

1877 \newenvironment{foreignabstract}{%
1878 \cleardoublepage
1879 \thispagestyle{plain}
1880 \begin{otherlanguage}{english}
1881 \foreignabstract@toptext\par
1882 \vspace*{8.6mm}
1883 \begin{center}
1884 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1885 \vspace*{13.2mm}
1886 \@authname\ \@authsurn \par
1887 \vspace*{7mm}
1888 \foreign@monthname/\number\year
1889 \end{center}\par
1890 \vspace*{1.5cm}
1891 \noindent%
1892 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1893 \foreign@advisorstring: &
1894 \count1=0
1895 \toks@={}
1896 \@whilenum \count1<\@advisor \do{%
1897 \ifcase\count1 % same as \ifnum0=\count1
1898 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1899 \expandafter\endcsname\expandafter\space%
1900 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1901 \else
1902 \toks@=\expandafter\expandafter\expandafter{%
1903 \expandafter\the\expandafter\toks@
1904 \expandafter&\expandafter\space
1905 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1906 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1907 }%
1908 \fi
1909 \advance\count1 by 1}
1910 \the\toks@
1911 \end{tabularx}\par
1912 \vspace*{2mm}
1913 \noindent\foreign@deptstring: \foreign@deptname\par
1914 \vspace*{7mm}}{%
1915 \end{otherlanguage}
1916 \vspace*{\fill}
1917 \global\let\@author\@empty
1918 \global\let\@date\@empty
1919 \global\let\foreign@title\@empty

```

```

1920 \global\let\foreign@title\relax
1921 \global\let\local@title\@empty
1922 \global\let\local@title\relax
1923 \global\let\author\relax
1924 \global\let\author\relax
1925 \global\let\date\relax}

```

\listoffigures

```

1926 \renewcommand\listoffigures{%
1927     \coppe@hasLof
1928     \if@twocolumn
1929         \@restonecoltrue\onecolumn
1930     \else
1931         \@restonecolfalse
1932     \fi
1933     \chapter*{\listfigurename}%
1934     \addcontentsline{toc}{chapter}{\listfigurename}%
1935     \@mkboth{\MakeUppercase\listfigurename}%
1936             {\MakeUppercase\listfigurename}%
1937     \@starttoc{lof}%
1938     \if@restonecol\twocolumn\fi
1939 }

```

\listoftables

```

1940 \renewcommand\listoftables{%
1941     \if@twocolumn
1942         \@restonecoltrue\onecolumn
1943     \else
1944         \@restonecolfalse
1945     \fi
1946     \chapter*{\listtablename}%
1947     \addcontentsline{toc}{chapter}{\listtablename}%
1948     \@mkboth{%
1949         \MakeUppercase\listtablename}%
1950         {\MakeUppercase\listtablename}%
1951     \@starttoc{lot}%
1952     \if@restonecol\twocolumn\fi
1953 }

```

\printlosymbols

```

1954 \newcommand\printlosymbols{%
1955 \renewcommand\glossaryname{\listsymbolname}%
1956 \@input@{\jobname.los}}

```

\makelosymbols

```

1957 \def\makelosymbols{%
1958     \newwrite\@losfile
1959     \immediate\openout\@losfile=\jobname.syx
1960     \newcommand\symb1[3] [] {\@bsphack\begin{group}
1961     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}}%
1962         \@sanitize%
1963         \@wrtlos{\@tempsymb1}{##2}{##3}\typeout%
1964     {Writing index of symbols file \jobname.syx}}%

```

```

1965 \let\makelosymbols\@empty%
1966 }%
1967 \@onlypreamble\makelosymbols

1968 \AtBeginDocument{%
1969 \@ifpackageloaded{hyperref}{%
1970 \newcommand\@wrls[3]{%
1971 \protected@write\@losfile{%
1972 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1973 \endgroup%
1974 \@esphack}}{%
1975 \newcommand\@wrls[3]{%
1976 \protected@write\@losfile{%
1977 {\string\indexentry{#1[#2] #3}{\thepage}}%
1978 \endgroup%
1979 \@esphack}}}%

\printloabbreviations

1980 \newcommand\printloabbreviations{%
1981 \renewcommand\glossaryname{\listabbreviationname}%
1982 \@input{\jobname.lab}}

\makeloabbreviations

1983 \def\makeloabbreviations{%
1984 \newwrite\@labfile
1985 \immediate\openout\@labfile=\jobname.abx
1986 \newcommand\abbrev[3][\@bsphack\beginingroup
1987 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
1988 \@sanitize
1989 \@wrlab{\@tempsymb1}{##2}{##3}}\typeout
1990 {Writing index of abbreviations file \jobname.abx}%
1991 \let\makeloabbreviations\@empty
1992 }
1993 \@onlypreamble\makeloabbreviations

1994 \AtBeginDocument{%
1995 \@ifpackageloaded{hyperref}{%
1996 \newcommand\@wrlab[3]{%
1997 \protected@write\@labfile{%
1998 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1999 \endgroup%
2000 \@esphack}}{%
2001 \newcommand\@wrlab[3]{%
2002 \protected@write\@labfile{%
2003 {\string\indexentry{#1[#2] #3}{\thepage}}%
2004 \endgroup%
2005 \@esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

2006 \newcommand\newsigla[3]{%

```

```

2007 \namedef{coppe@sigla@s@#1}{#2}%
2008 \namedef{coppe@sigla@l@#1}{#3}%
2009 \newcommand\coppe@sigla@reg[1]{%
2010 \ifundefined{abbrev}}{%
2011 {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
2012 \newcommand\coppe@sigla@full[1]{%
2013 \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}%
2014 \newcommand\coppe@sigla@undef[1]{%
2015 \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
2016 \textbf{??#1??}}
2017 \newcommand\coppe@sigla@first[1]{%
2018 \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
2019 \newcommand\coppe@sigla@norm[1]{%
2020 \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2021 {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
2022 {\@nameuse{coppe@sigla@s@#1}}}%
2023 \newcommand\coppe@sigla@star[1]{%
2024 \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2025 {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
2026 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

2027 %%% \AtBeginDocument{%
2028 %%% \ifpackageloaded{hyperref}{%
2029 %%% \def\@wrlab#1#2{%
2030 %%% \protected@write\@labfile{%
2031 %%% {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
2032 %%% \endgroup
2033 %%% \@esphack}}{%
2034 %%% \def\@wrlab#1#2{%
2035 %%% \protected@write\@labfile{%
2036 %%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
2037 %%% \endgroup
2038 %%% \@esphack}}}

2039 % Some macros used to generate cataloging information.
2040 \AtBeginDocument{%
2041 \ltx@ifpackageloaded{hyperref}{
2042 \def\coppe@bibEnd{%
2043 \immediate\write\@auxout{%
2044 \string\newlabel{bib:end}{\arabic{page}}{\page.\arabic{page}}}%
2045 \def\coppe@bibBegin{%
2046 \immediate\write\@auxout{%
2047 \string\newlabel{bib:begin}{\arabic{page}}{\page.\arabic{page}}}%
2048 \def\coppe@mainBegin{%
2049 \immediate\write\@auxout{%
2050 \string\newlabel{front:pageno}{\Roman{page}}{\page.\roman{page}}}%
2051 \def\coppe@hasLof{%
2052 \immediate\write\@auxout{%
2053 \string\newlabel{cat:lofflag}{\il.;}{\page.\roman{page}}}%
2054 }{%
2055 \def\coppe@bibEnd{%
2056 \immediate\write\@auxout{%
2057 \string\newlabel{bib:end}{\arabic{page}}}%
2058 \def\coppe@bibBegin{%
2059 \immediate\write\@auxout{%

```

```

2060     \string\newlabel{bib:begin}{\arabic{page}}{}}}%
2061     \def\coppe@mainBegin{%
2062         \immediate\write\@auxout{%
2063             \string\newlabel{front:pageno}{\Roman{page}}{}}}%
2064     \def\coppe@hasLof{%
2065         \immediate\write\@auxout{%
2066             \string\newlabel{cat:lofflag}{\il.}{}}}%
2067     }%
2068 }
2069 % The reference list is now produced by biblatex's \printbibliography (see the
2070 % bibliography setup after babel, above). The former thebibliography
2071 % redefinition and its \bibindent are no longer used and have been removed; the
2072 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
2073 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

2074 \newlength{\recuolongquote}%
2075 \setlength{\recuolongquote}{4cm}%
2076 \newenvironment*{longquote}[1][default]{%
2077     \list{}%
2078     \footnotesize%
2079     \addtolength{\leftskip}{\recuolongquote}%
2080     \item[]%
2081     \singlespacing%
2082     \ifthenelse{not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{}%
2083 }{\endlist}%

2084 \newenvironment{theglossary}{%
2085     \if@twocolumn%
2086         \@restonecoltrue\onecolumn%
2087     \else%
2088         \@restonecolfalse%
2089     \fi%
2090     \@mkboth{\MakeUppercase\glossaryname}%
2091     {\MakeUppercase\glossaryname}%
2092     \chapter*{\glossaryname}%
2093     \addcontentsline{toc}{chapter}{\glossaryname}%
2094     \list{}
2095     {\setlength{\listparindent}{0in}%
2096     \setlength{\labelwidth}{1.0in}%
2097     \setlength{\leftmargin}{1.5in}%
2098     \setlength{\labelsep}{0.5in}%
2099     \setlength{\itemindent}{0in}}%
2100     \sloppy}%
2101     {\if@restonecol\twocolumn\fi%
2102 \endlist}
2103 %
2104 \renewenvironment{theindex}{%
2105     \if@twocolumn%
2106         \@restonecolfalse%
2107     \else%
2108         \@restonecoltrue%
2109     \fi%
2110     \twocolumn[\@makeschapterhead{\indexname}]%

```

```

2111 \@mkboth{\MakeUppercase\indexname}%
2112 {\MakeUppercase\indexname}%
2113 \thispagestyle{plain}\parindent\z@
2114 \addcontentsline{toc}{chapter}{\indexname}
2115 \parskip\z@ \@plus .3\p@\relax
2116 \columnseprule \z@
2117 \columnsep 35\p@
2118 \let\item\@idxitem
2119 {\if@restonecol\onecolumn\else\clearpage\fi}
2120 \if@english
2121 \newcommand\listabbreviationname{List of Abbreviations and Acronyms}
2122 \newcommand\listsymbolname{List of Symbols}
2123 \newcommand\glossaryname{Glossary}
2124 \else
2125 \newcommand\listabbreviationname{Lista de Abreviaturas e Siglas}
2126 \newcommand\listsymbolname{Lista de S{\` i}mbolos}
2127 \newcommand\glossaryname{Gloss{\` a}rio}
2128 \fi
2129 %
2130 \newcommand\local@advisorstring{Orientador}
2131 \newcommand\foreign@advisorstring{Advisor}
2132 \ifthenelse{\boolean{maledoc}}{%
2133 \newcommand\local@approvedname{Examinado por}%
2134 }{%
2135 \newcommand\local@approvedname{Examinada por}%
2136 }
2137 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
2138 \newcommand\local@deptstring{Programa}
2139 \newcommand\foreign@deptstring{Department}
2140 \newcommand\local@cityname{Rio de Janeiro}
2141 \newcommand\local@statename{RJ}
2142 \newcommand\local@countryname{Brasil}
2143 %
2144 \newcommand\frontcover@maintext{
2145 \sloppy\nohyphens{\local@doctype\ de \@degreename\
2146 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2147 ao Programa de P{\` o}s-gradua{\c c}{\` a}o em \local@deptname,
2148 COPPE, da \local@universityname, como parte dos requisitos
2149 necess{\` a}rios {\` a} obten{\c c}{\` a}o do t{\` i}tulo de
2150 \local@degname\ em \local@deptname.}
2151 }
2152 %
2153 \newcommand\frontpage@maintext{
2154 \noindent {\MakeUppercase\local@doctype}
2155 \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
2156 \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
2157 DE P{\` O}S-GRADUA{\c C}{\` A}O E PESQUISA DE ENGENHARIA DA
2158 UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
2159 NECESS{\` A}RIOS PARA A OBTEN{\c C}{\` A}O DO GRAU DE
2160 {\MakeUppercase\local@degname} EM CI{\` E}NCIAS EM
2161 {\MakeUppercase\local@deptname.\par}}%
2162 }
2163 %
2164 \newcommand\frontpage@bottomtext{%

```

```

2165 \begin{center}
2166 {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
2167 {\MakeUppercase{\local@monthname\ DE \number\year}
2168 \end{center}%
2169 }
2170 %
2171 \newcommand\abstract@toptext{%
2172 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
2173 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2174 \sloppy\nohyphens{{\textquotesingle a} COPPE/UFRJ como parte dos requisitos
2175 necess{\textquotesingle a}rios para a obten{\c c}{\textasciitilde a}o do grau de
2176 \local@degname\ em Ci{\textasciitilde e}ncias (\@degree)}
2177 }
2178 \newcommand\foreignabstract@toptext{%
2179 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
2180 COPPE/UFRJ as a partial fulfillment of the requirements for the
2181 degree of \foreign@degname\ of Science (\@degree)}
2182 }
2183 %
2184 </class>
2185 <*glossary>
2186 actual '='
2187 quote '!'
2188 level '>'
2189 %%%% delim_0 " , p. "
2190 delim_0 "\\dotfill "
2191 lethead_flag 0
2192 headings_flag 0
2193 preamble
2194 "\n\\begin{theglossary}\n \\makeatletter"
2195 postamble
2196 "\n \\end{theglossary}\n"
2197 </glossary>
2198 <*class>
2199 \newcommand{\annex}{
2200 \iflanguage{brazilian}
2201 {\renewcommand{\appendixname}{Anexo}}
2202 {\renewcommand{\appendixname}{Annex}}
2203 \appendix
2204 \@coppeannextrue
2205 }
2206 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

2207 <*dbx>
2208 \ProvidesFile{coppe.dbx}[2026/05/23 v3.8 CoppeTeX ABNT datamodel]
2209 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
2210 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
2211 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
2212 % standard, music, audio, image, artwork, video, dataset, letter, periodical,

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2213 % performance) are already in biblatex's standard datamodel.
2214 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
2215 % Custom fields.
2216 \DeclareDatamodelFields[type=field,datatype=literal]{course}      % "(... em <course>)"
2217 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
2218 \DeclareDatamodelFields[type=field,datatype=literal]{platform}    % videogame platform
2219 \DeclareDatamodelFields[type=field,datatype=literal]{scale}       % map scale (escala)
2220 \DeclareDatamodelFields[type=list,datatype=literal]{artist}      % game/illustration art
2221 \DeclareDatamodelFields[type=list,datatype=literal]{director}    % movie/tvshow/video
2222 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
2223 \DeclareDatamodelFields[type=field,datatype=literal]{caderno}     % newspaper section
2224 \DeclareDatamodelFields[type=field,datatype=literal]{relator}     % jurisprudence rapporteur
2225 \DeclareDatamodelEntryfields{course,coppedegree}
2226 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
2227   {artist,platform,director,producer,version}
2228 \DeclareDatamodelEntryfields[map]{scale}
2229 \DeclareDatamodelEntryfields[article]{caderno}
2230 \DeclareDatamodelEntryfields[jurisdiction]{relator}
2231 </dbx>
2232 <*bbx>
2233 \ProvidesFile{coppe.bbx}[2026/05/23 v3.8 CoppeTeX ABNT bibliography style]
2234 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
2235 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
2236 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
2237 % so load setspace here as well (idempotent).
2238 \RequirePackage{setspace}
2239 % ABNT reference list keeps the date at the end (imprenta), same for author-date
2240 % and numeric modes, so we derive from the standard style, not authoryear.
2241 \RequireBibliographyStyle{standard}
2242
2243 % Note: standard style never dashes repeated authors, which already matches the
2244 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
2245 \ExecuteBibliographyOptions{%
2246   maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
2247   giveninits=false,
2248   uniquename=false,
2249   uniquelist=false,
2250   labeldateparts=true, % needed for author-date extradate (1999a/1999b)
2251   sorting=nyt,
2252 }
2253
2254 % Language strings (Disponível em / Acesso em / In, etc.)
2255 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
2256 \DeclareLanguageMapping{english}{english-coppe}
2257
2258 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
2259 \DeclareNameAlias{author}{family-given}
2260 \DeclareNameAlias{editor}{family-given}
2261 \DeclareNameAlias{translator}{family-given}
2262 \DeclareNameAlias{default}{family-given}
2263 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
2264 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
2265
2266 % Uppercase the surname (family + prefix) only inside the bibliography list;

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2267 % citations keep normal caps (post-2020 rule).
2268 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
2269 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
2270 % field names; the month stays abbreviated via \mkbibmonth.
2271 \protected\def\coppe@abntdate#1#2#3{%
2272   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
2273   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
2274   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}%
2275 \AtBeginBibliography{%
2276   \singlespacing % entries single-spaced internally (2.4)
2277   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
2278   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
2279   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
2280   \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
2281   \let\mkbibdateshort\coppe@abntdate
2282 }
2283
2284 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
2285 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
2286 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
2287 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
2288 % article AND the next word are uppercased; otherwise just the first word. The
2289 % delimited macros split off words; \coppeUCkeep strips the trailing-space
2290 % sentinel so no spurious space is left before the following punctuation.
2291 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
2292 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
2293   \coppeUCifart{#1}
2294   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
2295   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
2296 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
2297   \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
2298 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
2299 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
2300 \protected\def\coppeUCifart#1{%
2301   \begingroup
2302   \lowercase{\def\coppeUC@w{#1}}%
2303   \edef\coppeUC@w{\coppeUC@w,\coppeUC@w}%
2304   \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,}%
2305   \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
2306   \endgroup}
2307 \DeclareFieldFormat{title}{%
2308   \ifboolexpr{ test {\ifnameundef{labelname}} }
2309     {#1}%
2310     {\mkbibbold{#1}}%
2311 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
2312 % format receives the RAW title text (the title format above only sees the
2313 % field-printing commands), so \coppeUCtitle can split off and uppercase the
2314 % first word here.
2315 \DeclareFieldFormat{titlecase}{%
2316   \ifboolexpr{ test {\ifnameundef{labelname}} }
2317     {\coppeUCtitle{#1}}%
2318     {#1}}
2319 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
2320 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}

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2321 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
2322 % the title of a whole document, so override those back to bold.
2323 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
2324 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
2325 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
2326 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
2327 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
2328 \DeclareFieldFormat[booktitle]{\mkbibbold{#1}}
2329 \DeclareFieldFormat[maintitle]{\mkbibbold{#1}}
2330
2331 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
2332 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\addot}
2333 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\addot}
2334 \DeclareFieldFormat{pages}{p\addot\addnbspace#1}
2335
2336 % --- Periodical volume / number prefixes ---
2337 \DeclareFieldFormat[article]{volume}{v\addot\addnbspace#1}
2338 \DeclareFieldFormat[article]{number}{n\addot\addnbspace#1}
2339
2340 % --- "In:" before host title (incollection/inproceedings/inbook) ---
2341 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
2342
2343 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
2344 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
2345
2346 % --- ABNT article driver: no "In:; Journal, Local, v., n., p., date, DOI/URL.
2347 % Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
2348 % differ: the newspaper name (journaltitle) and city, then either
2349 % "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
2350 % "p. <pages>, <date>" (page before date, when none),
2351 % per NBR 6023 §4.2.3.6. ---
2352 \DeclareBibliographyDriver{article}{%
2353   \usebibmacro{bibindex}%
2354   \usebibmacro{begentry}%
2355   \usebibmacro{author/translator+others}%
2356   \setunit{\labelnamepunct}\newblock
2357   \usebibmacro{title}%
2358   \newunit\newblock
2359   \printfield{journaltitle}%
2360   \setunit{\addcomma\space}\printlist{location}%
2361   \iffieldequalstr{entrysubtype}{newspaper}
2362     {\iffieldundef{caderno}
2363       {\setunit{\addcomma\space}\printfield{pages}%
2364         \setunit{\addcomma\space}\printdate}%
2365       {\setunit{\addcomma\space}\printdate%
2366         \setunit{\addot\space}\printfield{caderno}%
2367         \setunit{\addcomma\space}\printfield{pages}}}%
2368   {\setunit{\addcomma\space}\printfield{volume}%
2369     \setunit{\addcomma\space}\printfield{number}%
2370     \setunit{\addcomma\space}\printfield{pages}%
2371     \setunit{\addcomma\space}\printdate}%
2372   \newunit\newblock
2373   \printfield{doi}%
2374   \newunit\newblock

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```

2375 \usebibmacro{url+urldate}%
2376 \usebibmacro{finentry}}
2377
2378 % --- incollection: ABNT puts the host organizer before the booktitle, as
2379 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
2380 % p. X-Y." The organizer role string is "Org./Orgs." (see the .ltx). ---
2381 \newbibmacro*{coppe:bookeditor}{%
2382 \ifnameundef{editor}%
2383 {}
2384 {\printnames{editor}%
2385 \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
2386 \clearname{editor}%
2387 \setunit{\adddot\space}}}%
2388 \newbibmacro*{coppe:partdriver}{%
2389 \usebibmacro{bibindex}\usebibmacro{begentry}%
2390 \usebibmacro{author/translator+others}%
2391 \setunit{\labelnamepunct}\newblock
2392 \usebibmacro{title}%
2393 \newunit\newblock
2394 \usebibmacro{in:}%
2395 \usebibmacro{coppe:bookeditor}%
2396 \printfield{booktitle}%
2397 \newunit\newblock
2398 \printfield{edition}%
2399 \newunit\newblock
2400 \printlist{location}%
2401 \setunit*{\addcolon\space}\printlist{publisher}%
2402 \setunit*{\addcomma\space}\printdate%
2403 \newunit\newblock
2404 \printfield{pages}%
2405 \setunit{\addcomma\space}\printfield{chapter}%
2406 \newunit\newblock
2407 \usebibmacro{url+urldate}%
2408 \usebibmacro{finentry}}
2409 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
2410 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
2411
2412 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
2413 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
2414
2415 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
2416 \DeclareFieldFormat{course}{\mkbibparens{\iflanguage{brazilian}{em\addspace}{in\addspace}#1}}
2417 \DeclareBibliographyDriver{thesis}{%
2418 \usebibmacro{bibindex}\usebibmacro{begentry}%
2419 \usebibmacro{author}%
2420 \setunit{\labelnamepunct}\newblock
2421 \usebibmacro{title}%
2422 \newunit\newblock
2423 \printfield{type}\setunit*{\addspace}\printfield{course}%
2424 \newunit\newblock
2425 \printlist{institution}%
2426 \setunit*{\addcomma\space}\printlist{location}%
2427 \setunit*{\addcomma\space}\printdate%
2428 \newunit\newblock

```

```

2429 \printfield{pagetotal}%
2430 \newunit\newblock
2431 \printfield{note}%
2432 \newunit\newblock
2433 \usebibmacro{url+urldate}%
2434 \usebibmacro{finentry}}
2435 % --- Games / audiovisual: render the custom fields (artist, platform,
2436 %      director, producer, version) the generic misc driver ignores. ---
2437 \newbibmacro*{coppe:media}{%
2438   \iflistundef{artist}{ }\setunit{\addperiod\space}%
2439   \printtext{\iflanguage{brazilian}{Ilustra\c{c}\~ao}{Art}\addcolon\space}\printlist{artist}%
2440   \iffieldundef{platform}{ }\setunit{\addperiod\space}%
2441   \printtext{\iflanguage{brazilian}{Plataforma}{Platform}\addcolon\space}\printfield{platform}%
2442   \iflistundef{director}{ }\setunit{\addperiod\space}%
2443   \printtext{\iflanguage{brazilian}{Dire\c{c}\~ao}{Directed by}\addcolon\space}\printlist{director}%
2444   \iflistundef{producer}{ }\setunit{\addperiod\space}%
2445   \printtext{\iflanguage{brazilian}{Produ\c{c}\~ao}{Produced by}\addcolon\space}\printlist{producer}%
2446 \newbibmacro*{coppe:mediadriver}{%
2447   \usebibmacro{bibindex}\usebibmacro{begentry}%
2448   \usebibmacro{author/translator+others}%
2449   \setunit{\labelnamepunct}\newblock
2450   \usebibmacro{title}%
2451   \newunit\newblock
2452   \usebibmacro{coppe:media}%
2453   \newunit\newblock
2454   \printlist{publisher}%
2455   \setunit*{\addcomma\space}\printlist{location}%
2456   \setunit*{\addcomma\space}\printdate%
2457   \newunit\newblock
2458   \printfield{version}%
2459   \newunit\newblock
2460   \printfield{note}%
2461   \newunit\newblock
2462   \usebibmacro{url+urldate}%
2463   \usebibmacro{finentry}}
2464 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
2465 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
2466 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
2467 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
2468 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
2469 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
2470 % imprenta) plus the mandatory scale ("Escala 1:N").
2471 \DeclareFieldFormat{scale}{\bibstring{coppescale}\adnbspace#1}
2472 \DeclareBibliographyDriver{map}{%
2473   \usebibmacro{bibindex}\usebibmacro{begentry}%
2474   \usebibmacro{author/translator+others}%
2475   \setunit{\labelnamepunct}\newblock
2476   \usebibmacro{title}%
2477   \newunit\newblock
2478   \printlist{location}%
2479   \setunit*{\addcolon\space}\printlist{publisher}%
2480   \setunit*{\addcomma\space}\printdate%
2481   \newunit\newblock
2482   \printfield{scale}%

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```

2483 \newunit\newblock
2484 \usebibmacro{url+urldate}%
2485 \usebibmacro{finentry}}
2486 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
2487 % number. date. The holder (depositante) is printed in normal case, not the
2488 % ALL-CAPS family treatment the bibliography gives the leading author.
2489 \newbibmacro*{coppe:depositor}{%
2490 \ifnameundef{holder}
2491 {}
2492 {\printtext{\bibstring{depositor}\addcolon\space}%
2493 \begingroup
2494 \def\mkbibnamefamily##1{##1}%
2495 \def\mkbibnameprefix##1{##1}%
2496 \printnames{holder}%
2497 \endgroup}}
2498 \DeclareBibliographyDriver{patent}{%
2499 \usebibmacro{bibindex}%
2500 \usebibmacro{begentry}%
2501 \usebibmacro{author/translator+others}%
2502 \setunit{\labelnamepunct}\newblock
2503 \usebibmacro{title}%
2504 \newunit\newblock
2505 \usebibmacro{coppe:depositor}%
2506 \newunit\newblock
2507 \printfield{type}%
2508 \setunit*{\addspace}\printfield{number}%
2509 \newunit\newblock
2510 \printdate%
2511 \newunit\newblock
2512 \usebibmacro{url+urldate}%
2513 \usebibmacro{finentry}}
2514 % Exotic ABNT types that biblatex's standard style leaves without a driver:
2515 % give each a baseline (book-like for published docs: norma/partitura; misc for
2516 % the rest) so they print. Per-type ABNT refinements can follow later.
2517 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
2518 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
2519 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
2520 % the law id + ementa (plain, not bold), then location and date.
2521 \newbibmacro*{coppe:legaldriver}{%
2522 \usebibmacro{bibindex}\usebibmacro{begentry}%
2523 \usebibmacro{author/translator+others}%
2524 \setunit{\labelnamepunct}\newblock
2525 \usebibmacro{title}%
2526 \newunit\newblock
2527 % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
2528 \iffieldundef{relator}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
2529 \printlist{location}%
2530 \setunit{\addcomma\space}\printdate%
2531 \newunit\newblock
2532 \usebibmacro{url+urldate}%
2533 \usebibmacro{finentry}}
2534 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
2535 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
2536 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}

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2537 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
2538 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
2539 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
2540 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
2541 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
2542 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
2543 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
2544
2545 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
2546 \DeclareSourcemap{%
2547   \maps[datatype=bibtex]{%
2548     % entry-type PT synonyms -> canonical English
2549     \map{\step[tourcesource=livro,typetarget=book]}
2550     \map{\step[tourcesource=artigo,typetarget=article]}
2551     \map{\step[tourcesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
2552     \map{\step[tourcesource=partedelivro,typetarget=incollection]}
2553     \map{\step[tourcesource=emlivro,typetarget=inbook]}
2554     \map{\step[tourcesource=verbete,typetarget=inreference]}
2555     \map{\step[tourcesource=anais,typetarget=proceedings]}
2556     \map{\step[tourcesource=trabalhodeevento,typetarget=inproceedings]}
2557     \map{\step[tourcesource=periodico,typetarget=periodical]}
2558     \map{\step[tourcesource=relatorio,typetarget=report]}
2559     \map{\step[tourcesource=relatoriotecnico,typetarget=report]}
2560     \map{\step[tourcesource>manual,typetarget>manual]}
2561     \map{\step[tourcesource=correspondencia,typetarget=letter]}
2562     \map{\step[tourcesource=patente,typetarget=patent]}
2563     \map{\step[tourcesource=diverso,typetarget=misc]}
2564     \map{\step[tourcesource=jogo,typetarget=game]}
2565     \map{\step[tourcesource=jogoeletronico,typetarget=videogame]}
2566     \map{\step[tourcesource=jogodetabuleiro,typetarget=boardgame]}
2567     \map{\step[tourcesource=filme,typetarget=movie]}
2568     \map{\step[tourcesource=serietv,typetarget=tvshow]}
2569     \map{\step[tourcesource=programadetelevisao,typetarget=tvshow]}
2570     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
2571     % except map (declared in coppe.dbx).
2572     \map{\step[tourcesource=legislacao,typetarget=legislation]}
2573     \map{\step[tourcesource=atonomativo,typetarget=legislation]}
2574     \map{\step[tourcesource=jurisprudencia,typetarget=jurisdiction]}
2575     \map{\step[tourcesource=documentojuridico,typetarget=legal]}
2576     \map{\step[tourcesource=documentocivil,typetarget=legal]}
2577     \map{\step[tourcesource=norma,typetarget=standard]}
2578     \map{\step[tourcesource=partitura,typetarget=music]}
2579     \map{\step[tourcesource=iconografico,typetarget=image]}
2580     \map{\step[tourcesource=obradearte,typetarget=artwork]}
2581     \map{\step[tourcesource=tridimensional,typetarget=artwork]}
2582     \map{\step[tourcesource=cartografico,typetarget=map]}
2583     \map{\step[tourcesource=mapa,typetarget=map]}
2584     \map{\step[tourcesource=audiovisual,typetarget=video]}
2585     \map{\step[tourcesource=conjuntodedados,typetarget=dataset]}
2586     \map{\step[tourcesource=fasciculo,typetarget=supperperiodical]}
2587     \map{\step[tourcesource=trabalhoapresentado,typetarget=unpublished]}
2588     % thesis family: retype to thesis and preset the (babel-aware) type word
2589     \map{\step[tourcesource=tese,typetarget=thesis]}
2590     \map{\step[tourcesource=dissertacao,typetarget=thesis]}

```

```

2591 \map{step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2592 \map{step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fieldv
2593 \map{step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
2594 \map{step[typesource=dscthesi,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
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2599 % field PT synonyms (unaccented) -> canonical English
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2622 \map{step[fieldsource=plataforma,fieldtarget=platform]}
2623 \map{step[fieldsource=diretor,fieldtarget=director]}
2624 \map{step[fieldsource=produtor,fieldtarget=producer]}
2625 \map{step[fieldsource=versao,fieldtarget=version]}
2626 \map{step[fieldsource=escala,fieldtarget=scale]}
2627 }%
2628 }
2629
2630 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
2631 \setlength{\bibhang}{0pt}
2632 \setlength{\bibitemsep}{\baselineskip}
2633
2634 </bbx>
2635 <*cbx>
2636 \ProvidesFile{coppe.cbx}[2026/05/23 v3.8 CoppeTeX ABNT author-date citations]
2637 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
2638 % author-year cites with normal-caps names (post-2020 rule) out of the box.
2639 \RequireCitationStyle{authoryear-comp}
2640
2641 % --- NBR 10520:2023 author separator -----
2642 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
2643 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
2644 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via

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2645 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
2646 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
2647 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
2648 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
2649 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
2650
2651 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
2652 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
2653 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
2654 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
2655 % so it is never added to the bibliography.
2656 % \citapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2657 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
2658 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2659 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
2660 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
2661 \DeclareRobustCommand{\citapud}[4][]{%
2662   #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
2663 \DeclareRobustCommand{\citepapud}[4][]{%
2664   \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
2665     \citeauthor{#4}\addcomma\space\citeyear{#4}%
2666     \ifblank{#1}{ }\addcomma\space\mkpageprefix[pagination]{#1}}}}
2667 </cbx>
2668 <*lxbxbr>
2669 \ProvidesFile{brazilian-coppe.lbx}[2026/05/23 v3.8 CoppeTeX brazilian strings]
2670 \InheritBibliographyExtras{brazilian}
2671 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
2672 \DeclareBibliographyStrings{%
2673   inherit      = {brazilian},
2674   availablefrom = {{Dispon\'}{\i}vel em}{Dispon\'}{\i}vel em}},
2675   urlseen      = {{Acesso em}{Acesso em}},
2676   in           = {{In}{In}},
2677   editor       = {{Organizador}{Org\adddot}},
2678   editors      = {{Organizadores}{Orgs\adddot}},
2679   depositor    = {{Depositante}{Depositante}},
2680   coppescale   = {{Escala}{Escala}},
2681   mscdiss      = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
2682   dscthesi     = {{Tese de Doutorado}{Tese de Doutorado}},
2683   phdthesis    = {{Tese de Doutorado}{Tese de Doutorado}},
2684   tcc          = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
2685 }
2686 </lxbxbr>
2687 <*lxbxen>
2688 \ProvidesFile{english-coppe.lbx}[2026/05/23 v3.8 CoppeTeX english strings]
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2691 \DeclareBibliographyStrings{%
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2694   urlseen      = {{Access on}{Access on}},
2695   in           = {{In}{In}},
2696   depositor    = {{Depositor}{Depositor}},
2697   coppescale   = {{Scale}{Scale}},
2698   mscdiss      = {{M\adddot Sc\adddot Dissertation}{M\adddot Sc\adddot Diss\adddot}},

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2699 dscthis = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
2700 phdthesis = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
2701 tcc = {{Undergraduate Thesis}{Undergraduate Thesis}},
2702 }
2703 </lboxen>
2704 <*numbbx>
2705 \ProvidesFile{coppe-numeric.bbx}[2026/05/24 v3.8 CoppeTeX ABNT numeric bibliography]
2706 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
2707 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
2708 % the PT->EN source map and the language mappings) and only adds the [n] list
2709 % label and the numeric sorting on top.
2710 \RequireBibliographyStyle{coppe}
2711
2712 % ABNT numeric system lists references in the order they are first cited
2713 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
2714 % numbered list. 'labelnumber' generates the [n] field.
2715 \ExecuteBibliographyOptions{labelnumber,sorting=none}
2716
2717 % Bracketed labels, both in the list and for any shorthand entries.
2718 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
2719 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
2720
2721 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
2722 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
2723 \defbibenvironment{bibliography}
2724 { \list
2725   {\printtext[labelnumberwidth]{%
2726     \printfield{labelprefix}%
2727     \printfield{labelnumber}}}
2728   {\setlength{\labelwidth}{\labelnumberwidth}%
2729     \setlength{\leftmargin}{\labelwidth}%
2730     \setlength{\labelsep}{\biblabelsep}%
2731     \addtolength{\leftmargin}{\labelsep}%
2732     \setlength{\itemsep}{\bibitemsep}%
2733     \setlength{\parsep}{\bibparsep}}%
2734     \renewcommand*{\makelabel}[1]{\hss#1}}
2735 {\endlist}
2736 {\item}
2737
2738 </numbbx>
2739 <*numcbx>
2740 \ProvidesFile{coppe-numeric.cbx}[2026/05/24 v3.8 CoppeTeX ABNT numeric citations]
2741 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
2742 % numbers [1], compressed and sorted within a single cite group. This is the
2743 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
2744 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
2745 % \citet -> Author [1].
2746 \RequireCitationStyle{numeric-comp}
2747 </numcbx>

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Acknowledgments

Thanks to all COPPE_{TE}X users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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Change History

v0.0		v2.2	
General: Creation Date.	1	General: Matching new guidelines, including new logo.	1
v0.1		v2.2.2	
General: Documentation: bibliography fixed, title translation.	1	General: Fixed some text constants in .bib and documented it here.	1
Sourceforge submission.	1	v3.0	
v0.2		General: Added support for monographs in English.	34
General: Unification of the code for the list of symbols and abbreviations.	1	New approval page layout.	40
v0.3		\department: Added new course on Nanotechnology.	35
General: Added ‘draft’ option.	1	\examiner: Examiners expansion without degree.	35
\maketitle: Added number of examiners test.	39	v3.1	
Generalization.	39	\department: Included a sort key in symb	35
v0.4		v3.2	
General: Beta documentation.	1	General: Fixed version problem between cls and dtx.	1
v0.5		v3.3	
abstract: Changed from macro to environment.	43	General: Extend abbrev to work like symb and accept a sorting key	1
\backmatter: Added mainmatter pages counter.	37	v3.4	
foreignabstract: Changed from macro to environment.	44	General: Some examples for figures, tables, longtables, etc.	1
v1.0		v3.4.1	
General: First COPPE _{TEX} release.	1	General: Fix english option to consider brazilian a second language, describe it in document	34
v2.0		v3.5	
General: COPPE _{TEX} release 2.0.	1	General: New command annex	59
v2.1		v3.5.1	
General: COPPE _{TEX} release 2.1: Matching the new rules.	1	General: New command annex now supports brazilian or english	59
\advisor: Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors.	35		